

Face Name Task Behavioural Study

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data and add labels

```
# Import data
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/BehaviourStudy/Behaviour")
data_raw <- read_csv("FNAT.csv")

# Assign appropriate Stim factors
data_raw$StimType <- factor(data_raw$StimType)
levels(data_raw$StimType) <- c("Sham", "TI 1:3")

# Assign factors
data_raw$StimType <- factor(data_raw$StimType)
data_raw$block <- factor(data_raw$block)
data_raw$S_ID <- factor(data_raw$S_ID)
data_raw$session <- factor(data_raw$session)
data_raw$Run <- factor(data_raw$Run)
```

Check data for outliers - Accuracy level

```
g_data <- ddply(data_raw, c("S_ID", "session", "Run"), summarise,
  N = sum(!is.na(trial)),
  mean_Acc_prct = (sum(Accuracy, na.rm=TRUE)/N)*100
)

pander(g_data)
```

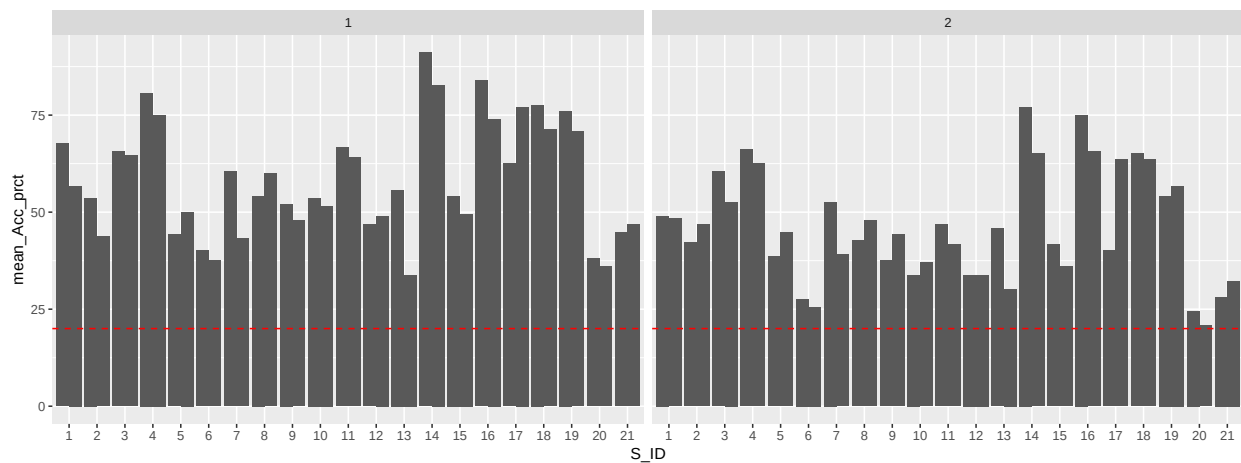
S_ID	session	Run	N	mean_Acc_prct
1	1	1	192	67.71
1	1	2	192	48.96

S_ID	session	Run	N	mean_Acc_prc
1	2	1	192	56.77
1	2	2	192	48.44
2	1	1	192	53.65
2	1	2	192	42.19
2	2	1	192	43.75
2	2	2	192	46.88
3	1	1	192	65.62
3	1	2	192	60.42
3	2	1	192	64.58
3	2	2	192	52.6
4	1	1	192	80.73
4	1	2	192	66.15
4	2	1	192	75
4	2	2	192	62.5
5	1	1	192	44.27
5	1	2	192	38.54
5	2	1	192	50
5	2	2	192	44.79
6	1	1	192	40.1
6	1	2	192	27.6
6	2	1	192	37.5
6	2	2	192	25.52
7	1	1	192	60.42
7	1	2	192	52.6
7	2	1	192	43.23
7	2	2	192	39.06
8	1	1	192	54.17
8	1	2	192	42.71
8	2	1	192	59.9
8	2	2	192	47.92
9	1	1	192	52.08
9	1	2	192	37.5
9	2	1	192	47.92
9	2	2	192	44.27
10	1	1	192	53.65
10	1	2	192	33.85
10	2	1	192	51.56
10	2	2	192	36.98
11	1	1	192	66.67
11	1	2	192	46.88
11	2	1	192	64.06
11	2	2	192	41.67
12	1	1	192	46.88
12	1	2	192	33.85
12	2	1	192	48.96
12	2	2	192	33.85
13	1	1	192	55.73
13	1	2	192	45.83
13	2	1	192	33.85
13	2	2	192	30.21
14	1	1	192	91.15
14	1	2	192	77.08

S_ID	session	Run	N	mean_Acc_prct
14	2	1	192	82.81
14	2	2	192	65.1
15	1	1	192	54.17
15	1	2	192	41.67
15	2	1	192	49.48
15	2	2	192	35.94
16	1	1	192	83.85
16	1	2	192	75
16	2	1	192	73.96
16	2	2	192	65.62
17	1	1	192	62.5
17	1	2	192	40.1
17	2	1	192	77.08
17	2	2	192	63.54
18	1	1	192	77.6
18	1	2	192	65.1
18	2	1	192	71.35
18	2	2	192	63.54
19	1	1	192	76.04
19	1	2	192	54.17
19	2	1	192	70.83
19	2	2	192	56.77
20	1	1	192	38.02
20	1	2	192	24.48
20	2	1	192	35.94
20	2	2	192	20.83
21	1	1	192	44.79
21	1	2	192	28.12
21	2	1	192	46.88
21	2	2	192	32.29

All subjects have mean accuracy above 20% (chance level)

```
ggplot(g_data, aes(x=S_ID, y=mean_Acc_prct, group=session)) +
  geom_bar(position=position_dodge(), stat="identity") + geom_hline(yintercept=20, linetype="dashed")
```



```
# Report values for immediate recall in methods - used as exclusion criteria
subset(g_data, Run == "1") %>% summarise(mean = mean(mean_Acc_prct),
  sd = sd(mean_Acc_prct), min = min(mean_Acc_prct), max = max(mean_Acc_prct))
```

```
##      mean      sd      min      max
## 1 58.45734 14.73355 33.85417 91.14583
```

```
# For delayed recall
subset(g_data, Run == "2") %>% summarise(mean = mean(mean_Acc_prct),
  sd = sd(mean_Acc_prct), min = min(mean_Acc_prct), max = max(mean_Acc_prct))
```

```
##      mean      sd      min      max
## 1 46.21776 14.16449 20.83333 77.08333
```

Check response distributions

```
## What is the distribution of responses (correct, foil, distractors, omission)
bdata <- data_raw
bdata$RespType <- ifelse(bdata$Accuracy==1, "Correct",
  ifelse(bdata$Foil==1, "Foil",
    ifelse(bdata$Distractor==1, "Distractor", NA)))
```

```
# ----- Checking number of cases (percent) -----
```

```
pander(bdata %>%
  dplyr::group_by(RespType, Run) %>%
  dplyr::summarise(mean = 100 * n() / (nrow(bdata)/2))) # 2 sessions
```

'summarise()' has grouped output by 'RespType'. You can override using the
'.groups' argument.

RespType	Run	mean
Correct	1	58.46
Correct	2	46.22
Distractor	1	12.29
Distractor	2	23.14
Foil	1	24.76
Foil	2	26.55
NA	1	4.489
NA	2	4.092

```
# Participants select Correct > Foil > Distractor. They are engaging with the task appropriately. NA =
```

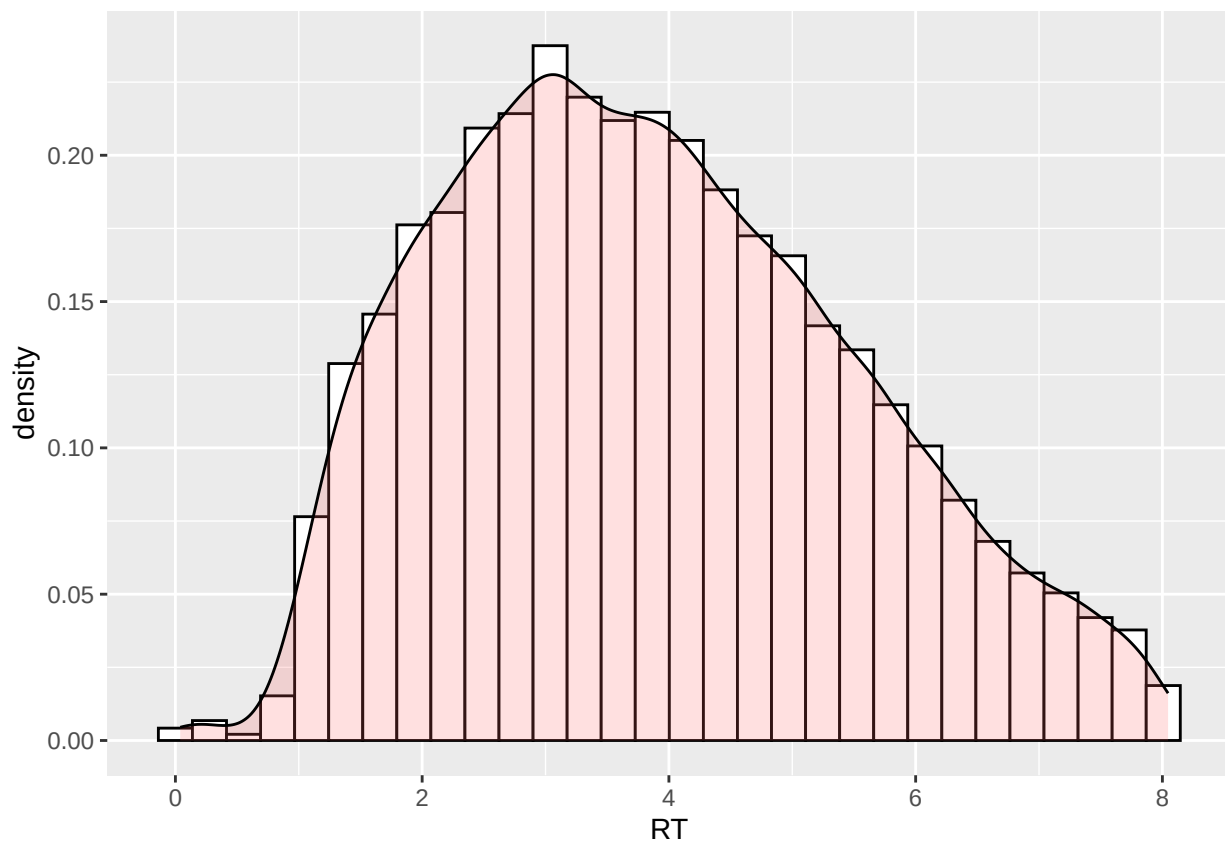
```
bdata <- bdata[!is.na(bdata$RespType),] # Getting rid of the few omissions in the dataset
bdata$RespType <- factor(bdata$RespType, levels = c("Correct", "Foil", "Distractor"))
```

Check the distribution of RTs and clean the data

```
# ---- Plot histogram of RT ----
ggplot(bdata, aes(x=RT)) +
  geom_histogram(aes(y=..density..), colour="black", fill="white")+
  geom_density(alpha=.2, fill="#FF6666")
```

```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(density)' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



```
# ---- Check 1 and 99% percentiles and remove those from the data ----
rts <- bdata$RT
qts <- quantile(rts, probs = c(.01, .99))
qts
```

```
##      1%      99%
## 1.011379 7.730183
```

```
#
#
bdata <- bdata %>% mutate(RT = replace(RT, RT<qts[1], NA))
bdata <- bdata %>% mutate(RT = replace(RT, RT>qts[2], NA))
bdata <- bdata[!is.na(bdata$RT),]

bdata <- bdata %>% mutate(RT = replace(RT, RT<0.3, NA))
bdata <- bdata[!is.na(bdata$RT),]
```

Add additional variables of interest

```
CorrectRT <- ifelse(bdata$Accuracy==1, bdata$RT, NA);
IncorrectRT <- ifelse(bdata$Accuracy==0, bdata$RT, NA);
CorrectConfidence <- ifelse(bdata$Accuracy==1, bdata$Confidence, NA);
IncorrectConfidence <- ifelse(bdata$Accuracy==0, bdata$Confidence, NA);

bdata <- cbind(bdata, CorrectRT, IncorrectRT, CorrectConfidence, IncorrectConfidence)
rm(CorrectRT,IncorrectRT,CorrectConfidence,IncorrectConfidence)
```

```
recall <- subset(bdata, Run == "1")
xtabs( ~ RespType + StimType, recall)
```

Accuracy - Recall

```
##           StimType
## RespType  Sham TI 1:3
##   Correct  2259  2381
##   Foil     1014   946
##   Distractor 499   470
```

```
g_bdata <- ddply(recall, c("S_ID","StimType", "RespType"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/192,
  meanRespTypePerc = meanRespType*100,
  medianRT = median(RT, na.rm=TRUE),
  meanRT = mean(RT, na.rm=TRUE),
  mean_Conf = mean(Confidence, na.rm=TRUE)
)

g_bdata <- g_bdata[!is.na(g_bdata$RespType),] # Getting rid of the few omissions in the dataset
g_bdata$RespType <- factor(g_bdata$RespType,levels = c("Correct", "Foil", "Distractor"))

### Means to report
g_bdata %>%
  group_by(StimType, RespType) %>%
  summarise_at(vars(meanRespTypePerc), list(mean = mean, std = sd))
```

```
## # A tibble: 6 x 4
## # Groups:   StimType [2]
##   StimType RespType    mean    std
##   <fct>    <fct>    <dbl> <dbl>
## 1 Sham      Correct     56.0  15.0
## 2 Sham      Foil        25.1   8.98
## 3 Sham      Distractor   12.4   8.48
## 4 TI 1:3    Correct     59.1  14.4
## 5 TI 1:3    Foil        23.5  10.0
## 6 TI 1:3    Distractor   11.7   7.02
```

```
# ----- Multinomial logistic regression -----
# Does the number of responses per response type vary for different stimulation conditions
recall$RespType <- relevel(recall$RespType, ref = "Correct")
m.acc <- nnet::multinom(RespType ~ StimType, data = recall)
```

```
## # weights: 9 (4 variable)
## initial value 8315.396413
## iter 10 value 6911.932070
## final value 6907.409400
## converged
```

```
summary(m.acc)
```

```
## Call:
## nnet::multinom(formula = RespType ~ StimType, data = recall)
##
## Coefficients:
##           (Intercept) StimTypeTI 1:3
## Foil          -0.8010193      -0.1220126
## Distractor    -1.5100683      -0.1124748
##
## Std. Errors:
##           (Intercept) StimTypeTI 1:3
## Foil           0.03780037      0.05390678
## Distractor     0.04946389      0.07067056
##
## Residual Deviance: 13814.82
## AIC: 13822.82
```

```
exp(coef(m.acc))
```

```
##           (Intercept) StimTypeTI 1:3
## Foil           0.4488712      0.8851372
## Distractor     0.2208949      0.8936199
```

```
# Check the Z-score for the model (wald Z)
z <- summary(m.acc)$coefficients/summary(m.acc)$standard.errors
z
```

```
##           (Intercept) StimTypeTI 1:3
## Foil          -21.19078     -2.263401
## Distractor    -30.52870     -1.591536
```

```
p <- (1 - pnorm(abs(z), 0, 1))*2
p
```

```
##           (Intercept) StimTypeTI 1:3
## Foil           0      0.02361101
## Distractor     0      0.11148897
```

```
Anova(m.acc)
```

```
## # weights:  6 (2 variable)
## initial value 8315.396413
## final value 6910.586023
## converged
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: RespType
##           LR Chisq Df Pr(>Chisq)
## StimType  6.3532  2   0.04173 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# ---- Group data for plotting ----
```

```
g_data <- ddply(recall, c("S_ID", "StimType", "session"), summarise,
  sum_acc = sum(Accuracy),
  sum_acc_perc = (sum_acc/192)*100,
  mean_acc = mean(Accuracy),
  sd_acc = sd(Accuracy),
  se_acc = se(Accuracy)
)
```

```
## Plot Accuracy
```

```
p.RespType <- ggplot(g_bdata, aes(x = RespType, y = meanRespTypePerc, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
scale_y_continuous(name = "Mean Response (%)",
  breaks = seq(0, 100, 20),
  limits=c(0, 100)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Correct", "Foil", "Distractor"),
    labels=c("Target", "Foil", "Distractor")) +
  theme(panel.grid.major = element_blank(),
```



```

    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 20),
          legend.text = element_text(size=14),
          legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("

```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

```

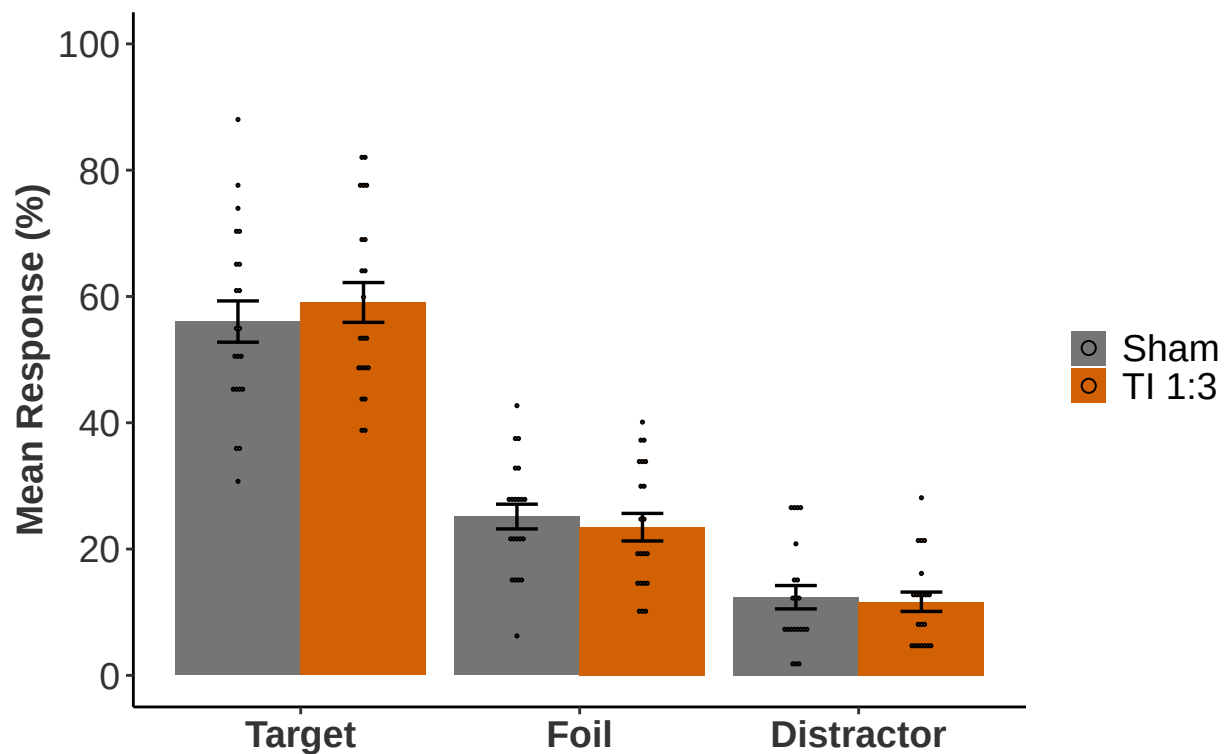
p.RespType

```

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

```



```

ggsave(p.RespType, filename = "Figure_5A.pdf", width = 11, height = 9, dpi = 300, units = "cm", device

```

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

```

```

# -----
#----- Binomial -----
m1 <- glmer(Accuracy ~ StimType + (1|S_ID) + (1|session), data=recall, family = binomial(link = "logit")
m2<- glmer(Accuracy ~ StimType + (1|S_ID) + (1|session) + (1|block), data=recall, family = binomial(link = "logit")
m3 <- glmer(Accuracy ~ StimType + (1|S_ID) + (1|session) + (1|block/trial), data=recall, family = binomial(link = "logit")

## boundary (singular) fit: see help('isSingular')

anova(m1, m2, m3)

## Data: recall
## Models:
## m1: Accuracy ~ StimType + (1 | S_ID) + (1 | session)
## m2: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block)
## m3: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##      npar    AIC    BIC logLik deviance  Chisq Df Pr(>Chisq)
## m1      4 9508.2 9535.9 -4750.1   9500.2
## m2      5 9452.7 9487.3 -4721.3   9442.7 57.513  1 3.358e-14 ***
## m3      6 9454.7 9496.2 -4721.3   9442.7  0.000  1      1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

lmtest::lrtest(m1, m2, m3)

## Likelihood ratio test
##
## Model 1: Accuracy ~ StimType + (1 | S_ID) + (1 | session)
## Model 2: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block)
## Model 3: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##      #Df LogLik Df  Chisq Pr(>Chisq)
## 1      4 -4750.1
## 2      5 -4721.3  1 57.513 3.358e-14 ***
## 3      6 -4721.3  1  0.000      1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# keeping m2

Anova(m2)

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: Accuracy
##              Chisq Df Pr(>Chisq)
## StimType 5.8566  1  0.01552 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
summary(m2)
```

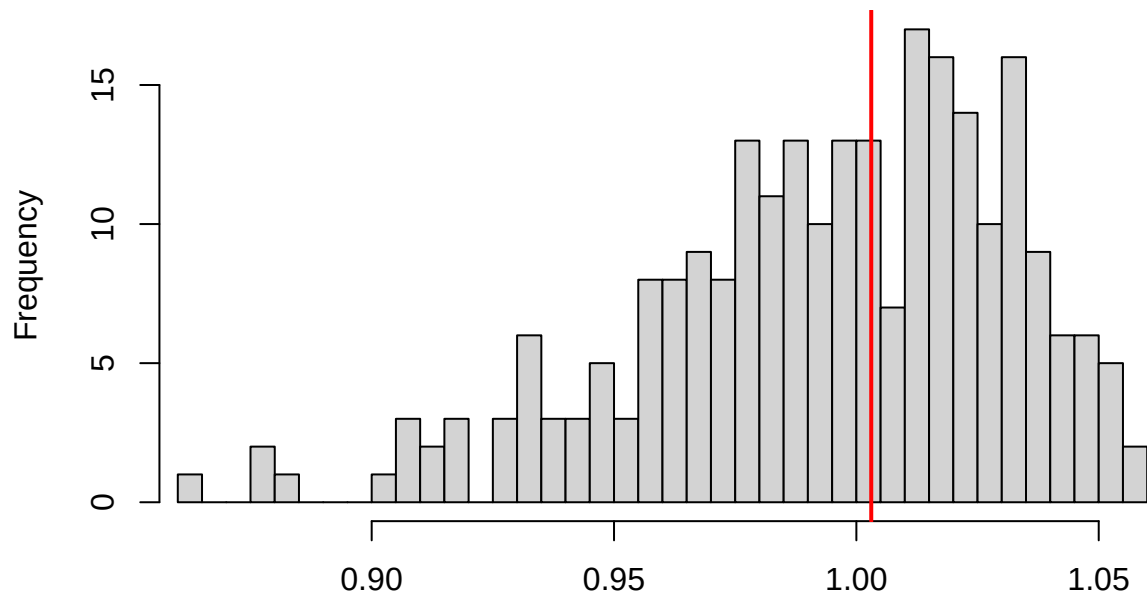
```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block)
## Data: recall
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##      AIC      BIC   logLik deviance df.resid
##  9452.7   9487.3  -4721.3   9442.7     7564
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4124 -0.9582  0.5097  0.7853  1.6233
##
## Random effects:
## Groups Name          Variance Std.Dev.
## S_ID   (Intercept) 0.43737  0.6613
## block  (Intercept) 0.05483  0.2342
## session (Intercept) 0.01449  0.1204
## Number of obs: 7569, groups: S_ID, 21; block, 12; session, 2
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.46260    0.18411   2.513   0.0120 *
## StimTypeTI 1:3  0.12083    0.04993   2.420   0.0155 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## StmTypTI1:3 -0.135
```

```
library(DHARMA)
```

```
## This is DHARMA 0.4.5. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')
```

```
testDispersion(m2)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**

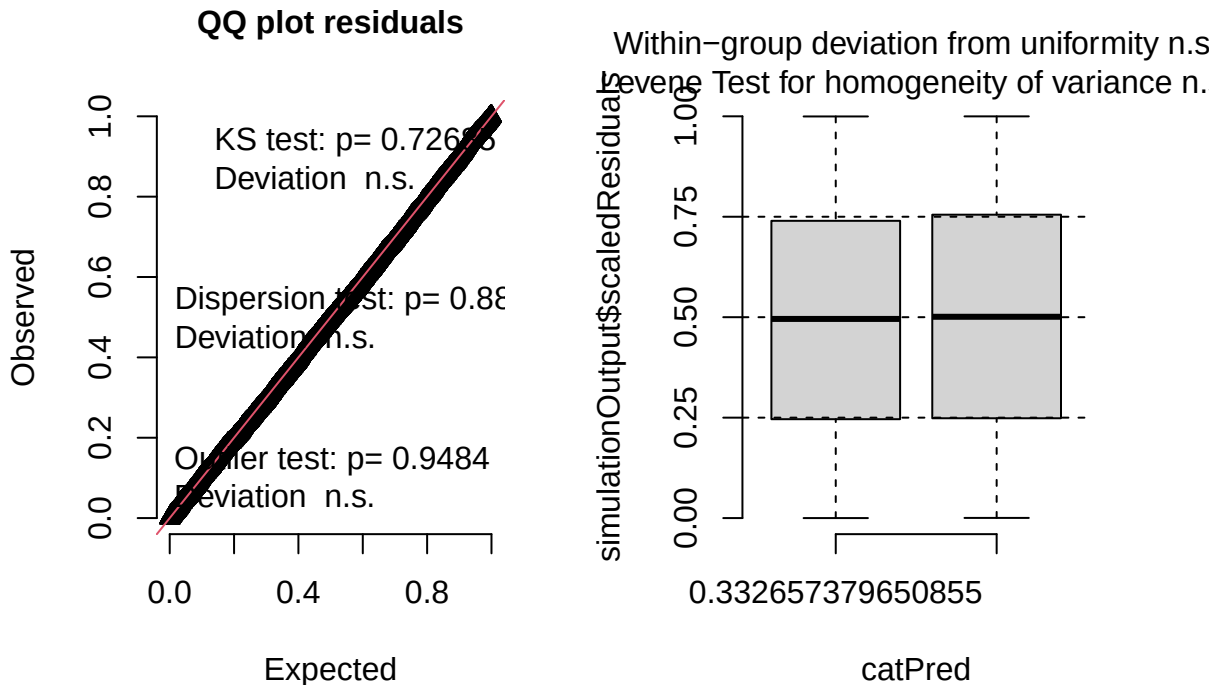


Simulated values, red line = fitted model. p-value (two.sided) = 0.888

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0097, p-value = 0.888
## alternative hypothesis: two.sided
```

```
simulationOutput <- simulateResiduals(fittedModel = m2, plot = T)
```

DHARMA residual



```
emm2 = emmeans(m2, specs = "StimType")
emm2
```

```
## StimType emmean SE df asymp.LCL asymp.UCL
## Sham 0.463 0.184 Inf 0.102 0.823
## TI 1:3 0.583 0.184 Inf 0.223 0.944
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
emmeans(m2, pairwise ~ StimType, type="response")
```

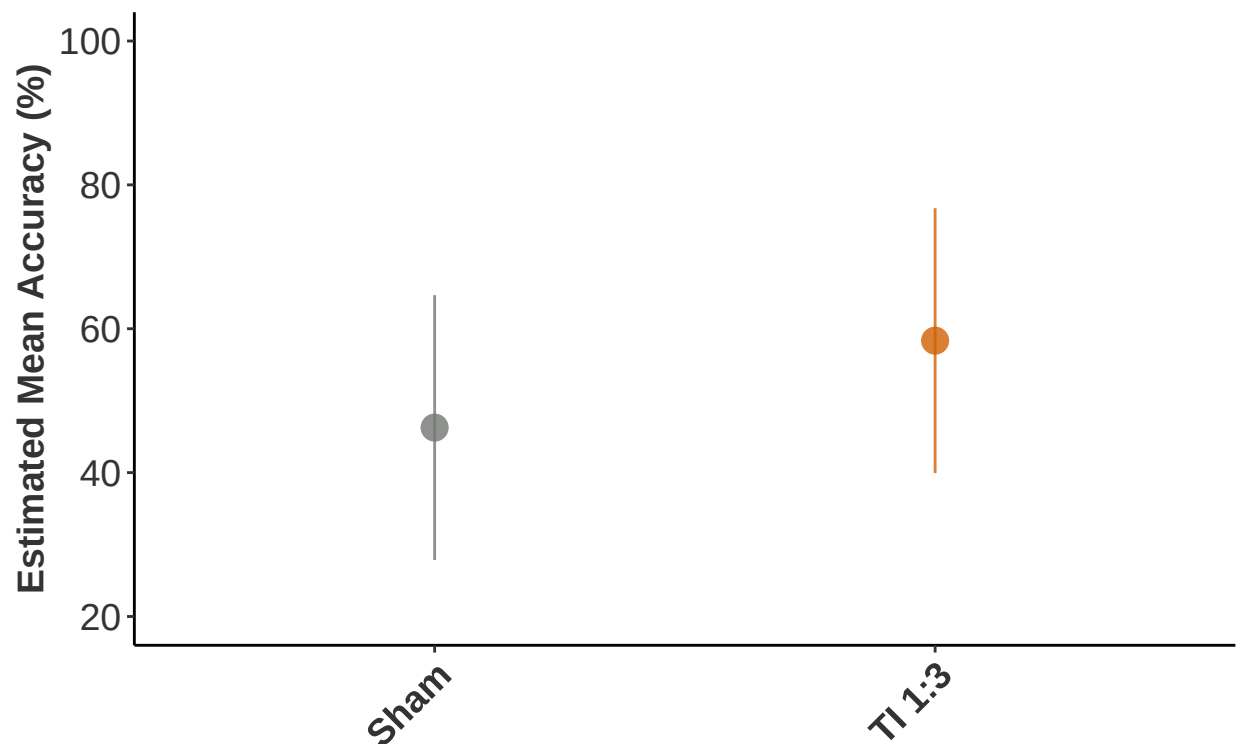
```
## $emmeans
## StimType prob SE df asymp.LCL asymp.UCL
## Sham 0.614 0.0437 Inf 0.525 0.695
## TI 1:3 0.642 0.0423 Inf 0.555 0.720
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
##
## $contrasts
## contrast odds.ratio SE df null z.ratio p.value
## Sham / TI 1:3 0.886 0.0442 Inf 1 -2.420 0.0155
##
## Tests are performed on the log odds ratio scale
```

```
pairs(emm2, reverse = TRUE, type="response")
```

```
## contrast      odds.ratio      SE df null z.ratio p.value
## TI 1:3 / Sham      1.13 0.0563 Inf  1  2.420  0.0155
##
## Tests are performed on the log odds ratio scale
```

```
fit <- as.data.frame(emm2)
g_data <- ddply(recall, c("S_ID", "StimType"), summarise,
                mean_acc = mean(Accuracy))

## Plot Accuracy - binomial
p <- ggplot(data = fit, mapping=aes(x=StimType, y=emmean*100, ymin=(emmean-SE)*100, ymax=(emmean+SE)*100)) +
  geom_pointrange(size=0.9, alpha=0.8, color= c("#747674", "#D16103")) +
  scale_y_continuous(name = "Estimated Mean Accuracy (%)",
                    limits=c(20, 100)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1),
        axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
        axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        panel.background = element_blank(),
        legend.title=element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 20)) +
  ggtitle("") + xlab("") + scale_color_manual(values = c("#747674", "#D16103"), name = "", labels = c(
p
```



```
ggsave(p, filename = "Figure_5b.pdf", width = 4, height = 9, dpi = 300, units = "cm", device = "pdf")
```

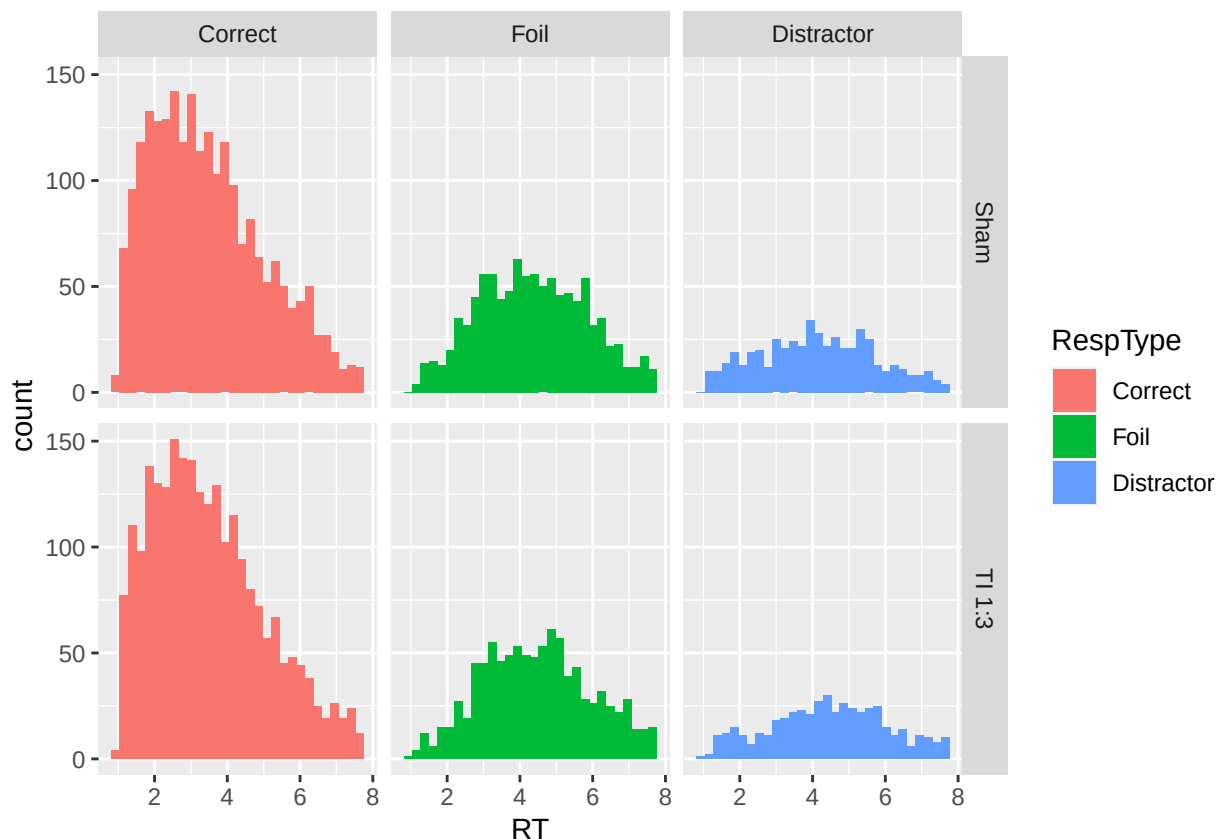
```
# plotting the data
library(ggribes)
```

Reaction Times - Recall

```
##
## Attaching package: 'ggribes'

## The following objects are masked from 'package:tidybayes':
##
##   scale_point_color_continuous, scale_point_color_discrete,
##   scale_point_colour_continuous, scale_point_colour_discrete,
##   scale_point_fill_continuous, scale_point_fill_discrete,
##   scale_point_size_continuous
```

```
ggplot(recall, aes(x = RT, fill = RespType)) +
  geom_histogram(bins = 30) +
  facet_grid(StimType ~ RespType)
```



```
pander(recall %>%
  group_by(RespType, StimType) %>%
  summarise(median(RT), sd(RT)))
```

'summarise()' has grouped output by 'RespType'. You can override using the
'.groups' argument.

RespType	StimType	median(RT)	sd(RT)
Correct	Sham	3.22	1.538
Correct	TI 1:3	3.252	1.544
Foil	Sham	4.326	1.475
Foil	TI 1:3	4.443	1.484
Distractor	Sham	4.1	1.58
Distractor	TI 1:3	4.433	1.597

```
# Does reaction time per response type vary for different stimulation conditions?

# Using glm without random effects to find starting values for the optimisation

m0 <- glm(RT ~ StimType*RespType, data=recall, family = inverse.gaussian(link=identity))

m1 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|session),
  data = recall,
```



```

    family = inverse.gaussian(link=identity),
    start=list(fixef=coef(m0)),
    control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m2 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|block) + (1|session),
    data = recall,
    family = inverse.gaussian(link=identity),
    start=list(fixef=coef(m0)),
    control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m3 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|session) + (1|block/trial),
    data = recall,
    family = inverse.gaussian(link=identity),
    start=list(fixef=coef(m0)),
    control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

anova(m1, m2, m3)

```

```

## Data: recall
## Models:
## m1: RT ~ StimType * RespType + (1 | S_ID) + (1 | session)
## m2: RT ~ StimType * RespType + (1 | S_ID) + (1 | block) + (1 | session)
## m3: RT ~ StimType * RespType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##      npar   AIC    BIC logLik deviance   Chisq Df Pr(>Chisq)
## m1      9 27101 27164 -13542   27083
## m2     10 27097 27167 -13539   27077  5.9272  1    0.01491 *
## m3     11 27048 27124 -13513   27026 51.4553  1    7.325e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

lmtest::lrtest(m1, m2, m3)

```

```

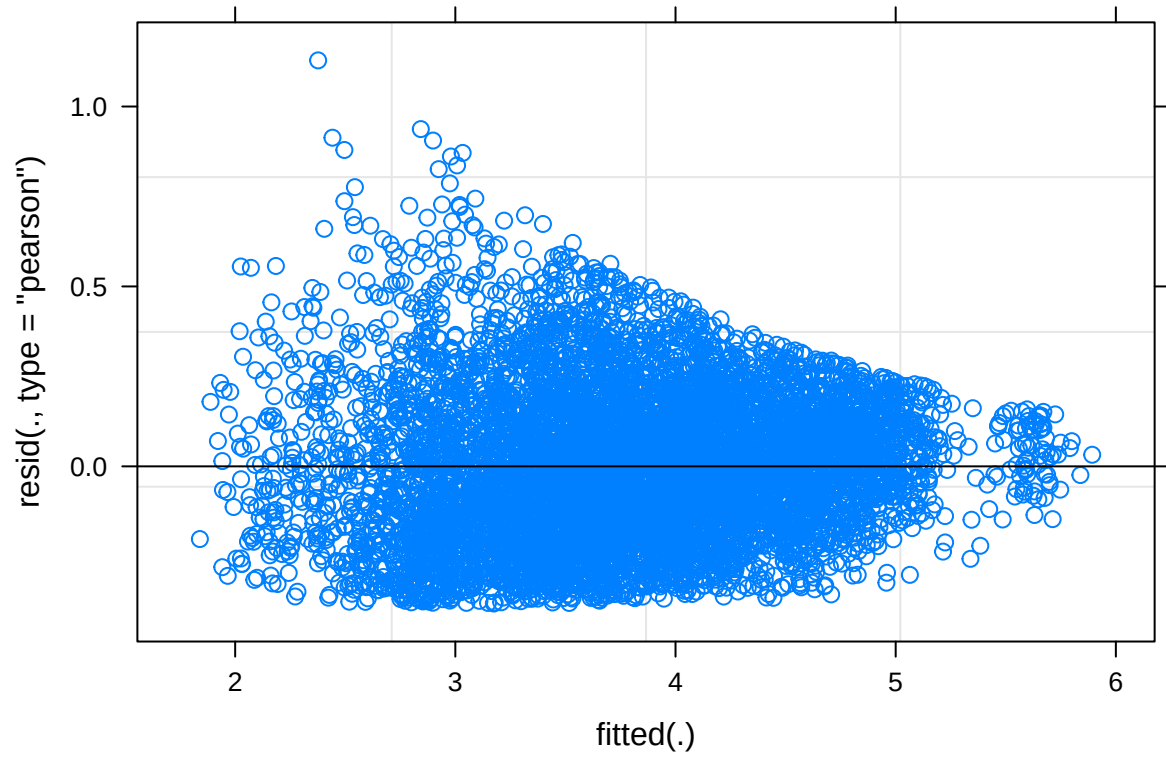
## Likelihood ratio test
##
## Model 1: RT ~ StimType * RespType + (1 | S_ID) + (1 | session)
## Model 2: RT ~ StimType * RespType + (1 | S_ID) + (1 | block) + (1 | session)
## Model 3: RT ~ StimType * RespType + (1 | S_ID) + (1 | session) + (1 |
##      block/trial)
##      #Df LogLik Df   Chisq Pr(>Chisq)
## 1      9 -13542
## 2     10 -13539  1  5.9272    0.01491 *
## 3     11 -13513  1 51.4553   7.325e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

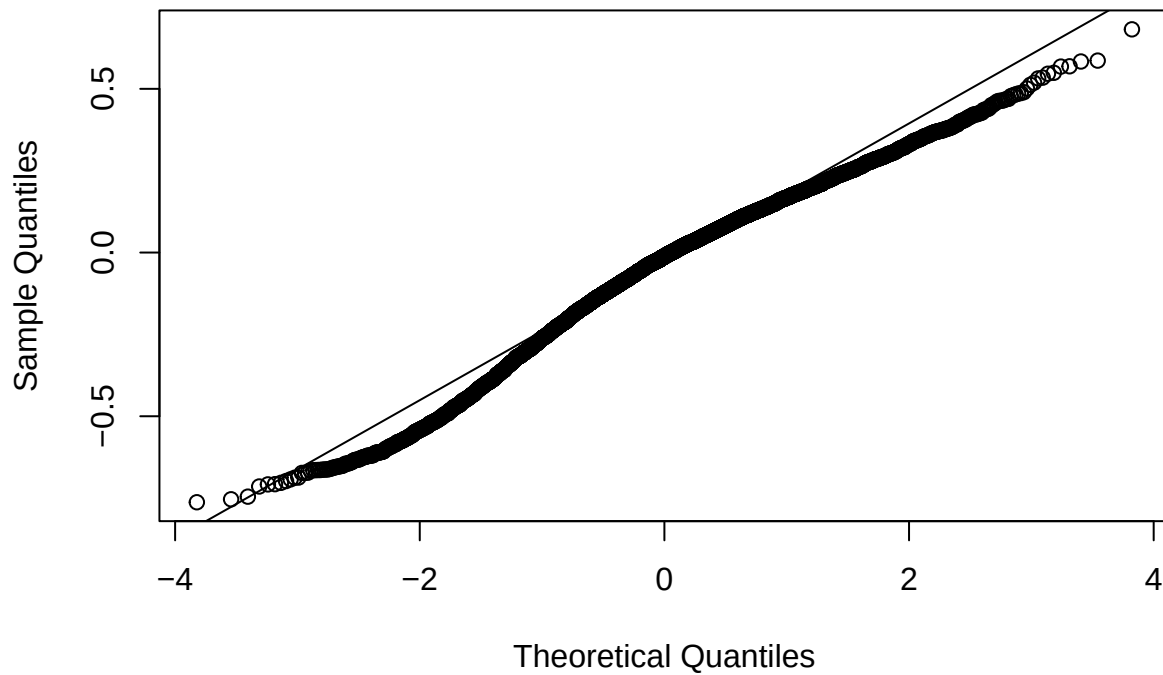
# keeping m3
plot(m3)

```



```
qqnorm(resid(m3))  
qqline(resid(m3))
```

Normal Q-Q Plot



```
Anova(m3)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##               Chisq Df Pr(>Chisq)
## StimType       3.0172  1    0.08239 .
## RespType      424.7137  2    < 2e-16 ***
## StimType:RespType  2.8268  2    0.24332
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m3)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
## Family: inverse.gaussian ( identity )
## Formula: RT ~ StimType * RespType + (1 | S_ID) + (1 | session) + (1 |
##   block/trial)
## Data: recall
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
##   nAGQ0initStep = FALSE)
##
##      AIC      BIC    logLik deviance df.resid
## 27047.8 27124.1 -13512.9 27025.8     7558
```

```
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8856 -0.7156 -0.0577  0.6329  5.5816
##
## Random effects:
##      Groups       Name             Variance Std.Dev.
## trial:block (Intercept) 0.027405  0.16555
## S_ID        (Intercept) 0.061743  0.24848
## block       (Intercept) 0.000925  0.03041
## session     (Intercept) 0.002962  0.05443
## Residual                    0.040860  0.20214
## Number of obs: 7569, groups: trial:block, 192; S_ID, 21; block, 12; session, 2
##
## Fixed effects:
##                                     Estimate Std. Error t value Pr(>|z|)
## (Intercept)                        3.74713     0.24366  15.379   <2e-16 ***
## StimTypeTI 1:3                      0.03973     0.04108   0.967   0.3335
## RespTypeFoil                       0.84082     0.06506  12.924   <2e-16 ***
## RespTypeDistractor                 0.71878     0.08008   8.976   <2e-16 ***
## StimTypeTI 1:3:RespTypeFoil         0.03812     0.09580   0.398   0.6907
## StimTypeTI 1:3:RespTypeDistractor  0.20520     0.12279   1.671   0.0947 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) StTTI1:3 RspTyF RspTyD STTI1:3:RTF
## StmTypTI1:3 -0.102
## RespTypeFol -0.052  0.278
## RspTypDstrc -0.038  0.215   0.156
## STTI1:3:RTF  0.033 -0.396  -0.667 -0.094
## STTI1:3:RTD  0.023 -0.300  -0.093 -0.644  0.128
```

```
estimated_means_RTall <- emmeans(m3, pairwise ~ StimType | RespType, type="response")
```

```
# -- Does reaction time for correct vs incorrect vary for different stimulation conditions
m0 <- glm(RT ~ StimType*Accuracy, data=recall, family = inverse.gaussian(link=identity))
```

```
m2 <- glmer(RT ~ StimType*Accuracy + (1|S_ID) + (1|session) + (1|block),
  data = recall,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))
```

```
m3 <- glmer(RT ~ StimType*Accuracy + (1|S_ID) + (1|session) + (1|block/trial),
  data = recall,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))
```

```
anova(m0, m2, m3)
```

```
## Analysis of Deviance Table
##
## Model: inverse.gaussian, link: identity
##
## Response: RT
##
## Terms added sequentially (first to last)
##
##
##           Df Deviance Resid. Df Resid. Dev
## NULL                7568      459.30
## StimType           1   0.1048      7567      459.19
## Accuracy            1  25.1659      7566      434.03
## StimType:Accuracy   1   0.1313      7565      433.90
```

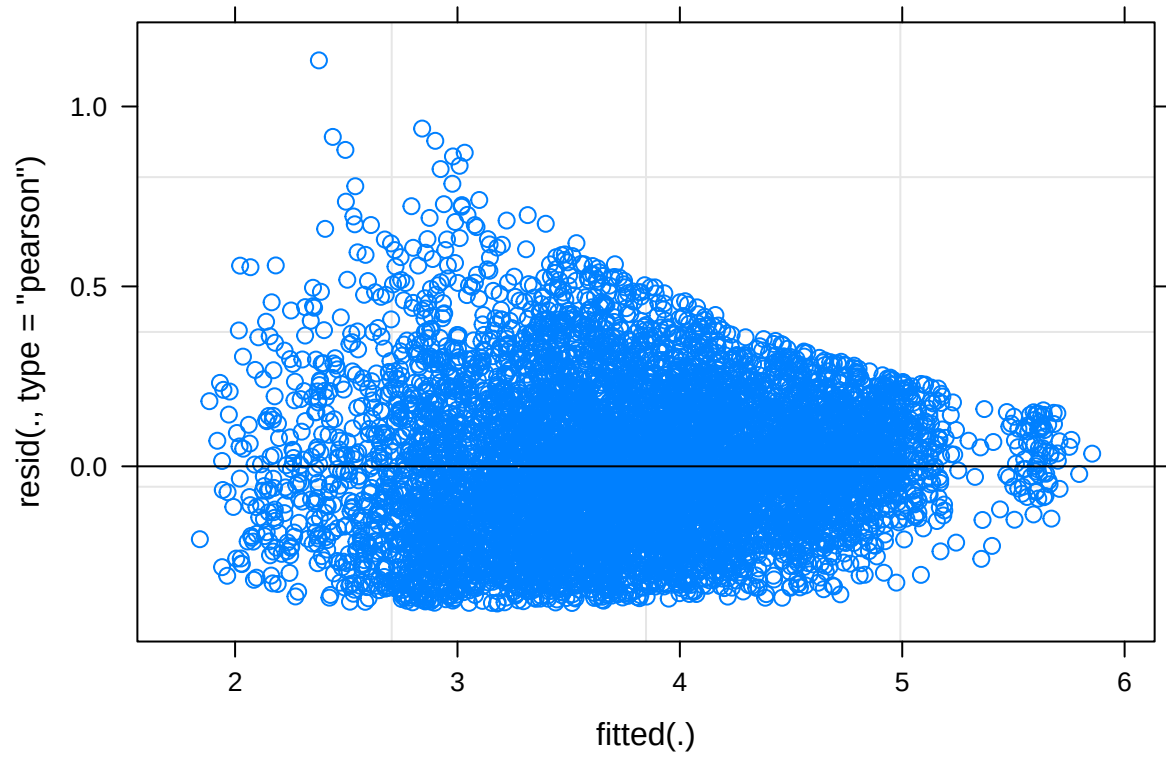
```
lmtest::lrtest(m0, m2, m3)
```

```
## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"
```

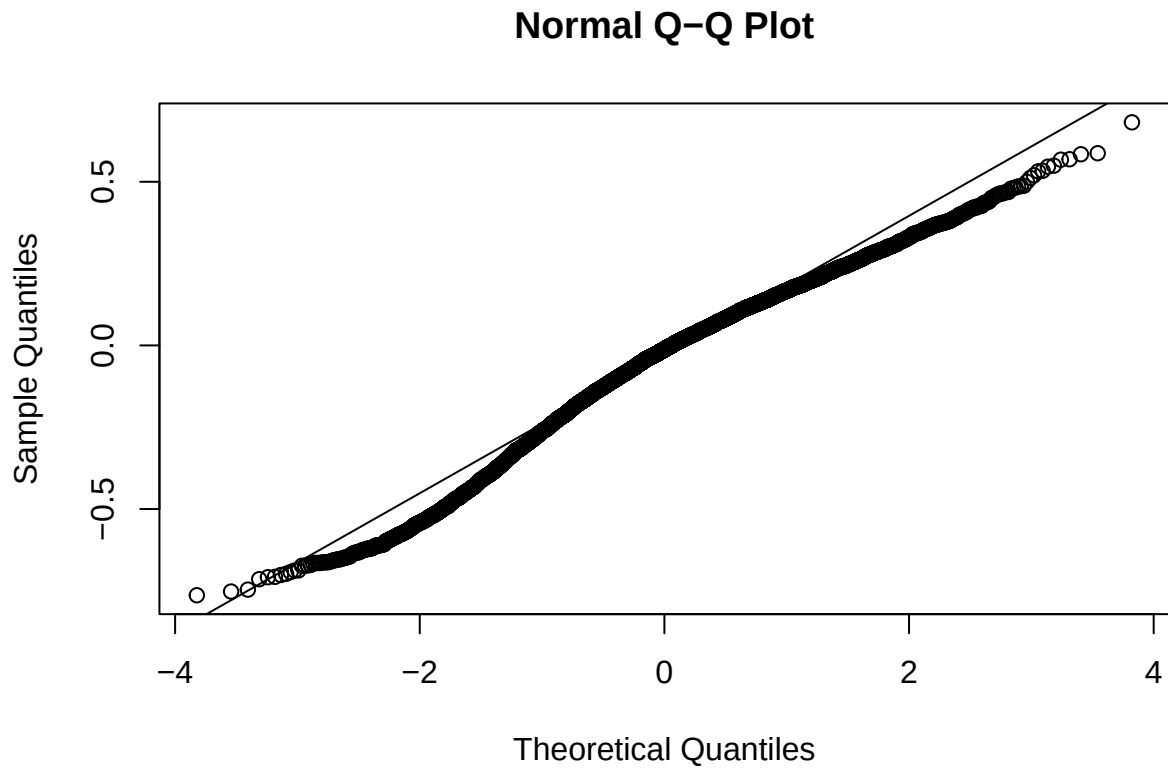
```
## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"
```

```
## Likelihood ratio test
##
## Model 1: RT ~ StimType * Accuracy
## Model 2: RT ~ StimType * Accuracy + (1 | S_ID) + (1 | session) + (1 |
##      block)
## Model 3: RT ~ StimType * Accuracy + (1 | S_ID) + (1 | session) + (1 |
##      block/trial)
##   #Df LogLik Df    Chisq Pr(>Chisq)
## 1    5 -14041
## 2    8 -13540  3 1003.163 < 2.2e-16 ***
## 3    9 -13514  1   51.294 7.952e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(m3)
```



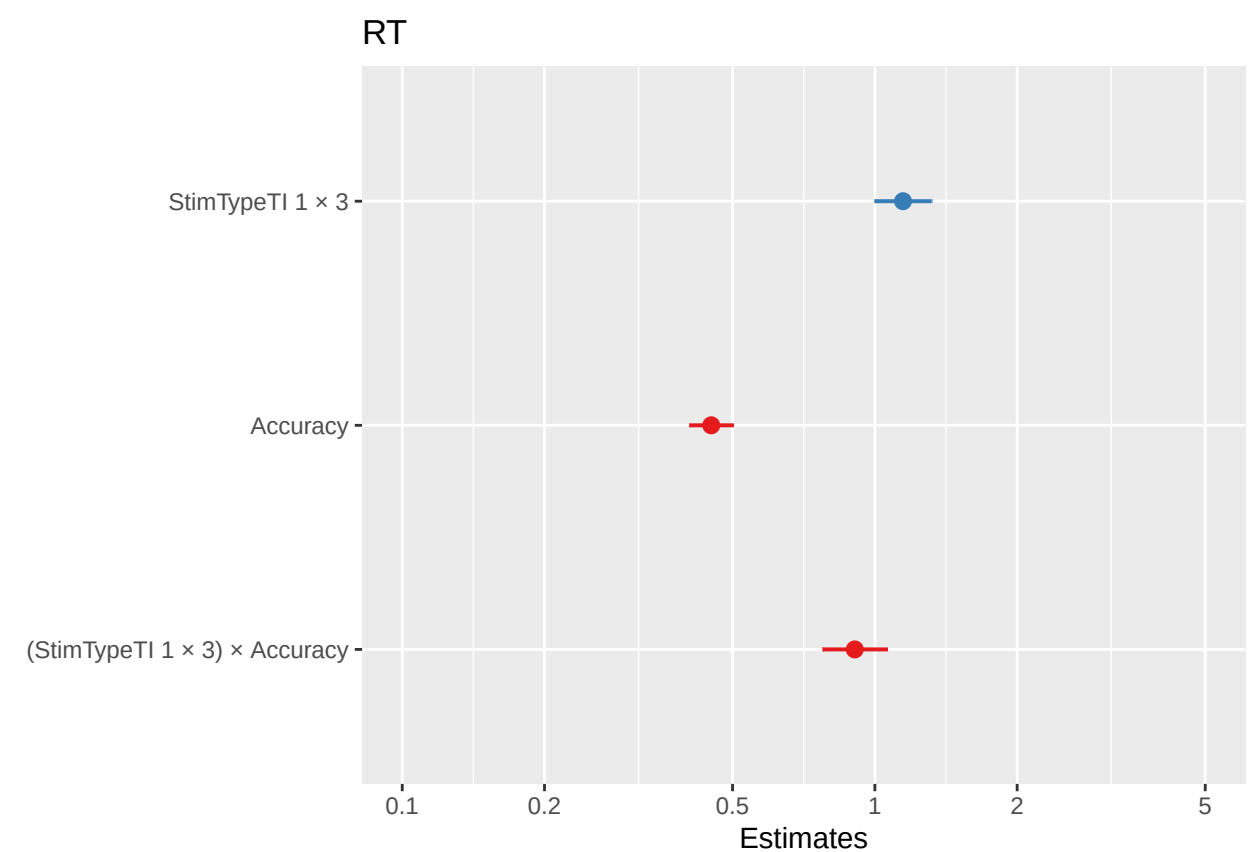
```
qqnorm(resid(m3))  
qqline(resid(m3))
```



```
Anova(m3)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##               Chisq Df Pr(>Chisq)
## StimType       2.9929  1    0.08363 .
## Accuracy      425.4772  1    < 2e-16 ***
## StimType:Accuracy  1.4947  1    0.22149
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot_model(m3)
```



```
pander(recall %>%
  group_by(Accuracy, StimType) %>%
  summarise(median(RT), sd(RT)))
```

'summarise()' has grouped output by 'Accuracy'. You can override using the
'.groups' argument.

Accuracy	StimType	median(RT)	sd(RT)
0	Sham	4.258	1.513
0	TI 1:3	4.437	1.522
1	Sham	3.22	1.538
1	TI 1:3	3.252	1.544

```
# Check RT for correct Trials only -
#
recall_correct <- subset(recall, Accuracy=="1")

m0 <- glm(RT ~ StimType, data=recall_correct, family = inverse.gaussian(link=identity))

m2 <- glmer(RT ~ StimType + (1|S_ID) + (1|block) + (1|session),
  data = recall_correct,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
```



```

        control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))
m3 <- glmer(RT ~ StimType + (1|S_ID) + (1|session) + (1|block/trial),
            data = recall_correct,
            family = inverse.gaussian(link=identity),
            start=list(fixef=coef(m0)),
            control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

```

```
## boundary (singular) fit: see help('isSingular')
```

```
anova(m0, m2, m3)
```

```
## Analysis of Deviance Table
##
## Model: inverse.gaussian, link: identity
##
## Response: RT
##
## Terms added sequentially (first to last)
##
##
```

	Df	Deviance	Resid. Df	Resid. Dev
## NULL			4639	320.06
## StimType	1	0.019676	4638	320.04

```
lmtest::lrtest(m0, m2, m3)
```

```
## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"
```

```
## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"
```

```
## Likelihood ratio test
##
## Model 1: RT ~ StimType
## Model 2: RT ~ StimType + (1 | S_ID) + (1 | block) + (1 | session)
## Model 3: RT ~ StimType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##   #Df  LogLik Df   Chisq Pr(>Chisq)
## 1    3 -8326.5
## 2    6 -8075.8  3 501.318 < 2.2e-16 ***
## 3    7 -8055.3  1  40.987 1.532e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(m2)
```

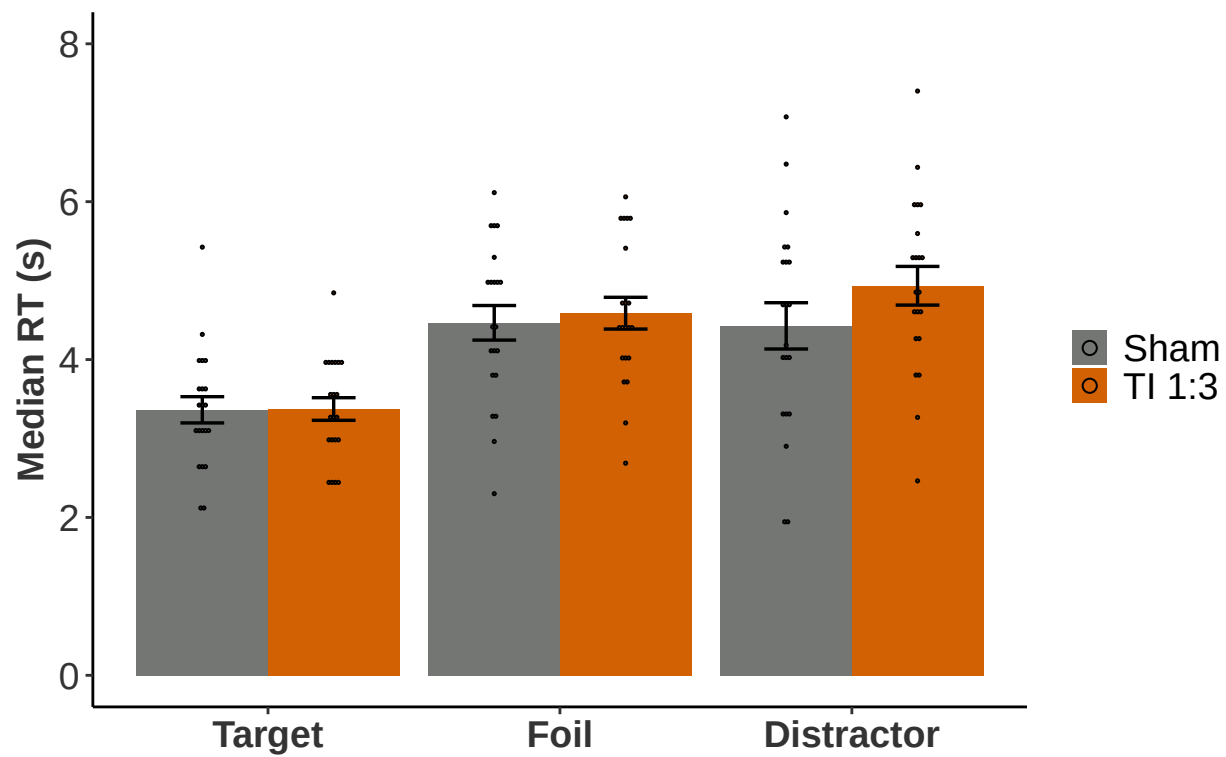
```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##           Chisq Df Pr(>Chisq)
## StimType 1.4906  1    0.2221
```

```
## Plot RT
```

```
p.RespTypeRT <- ggplot(g_bdata, aes(x = RespType, y = medianRT, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.9) +
  scale_y_continuous(name = "Median RT (s)",
    breaks = seq(0, 8, 2),
    limits=c(0, 8)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Correct","Foil", "Distractor"),
    labels=c("Target","Foil", "Distractor")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size=14),
    legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("Correct" = "#1f77b4", "Foil" = "#ff7f0e", "Distractor" = "#2ca02c"))

p.RespTypeRT
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.
```



```
ggsave(p.RespTypeRT, filename = "Figure_5D.pdf", width = 11, height = 9, dpi = 300, units = "cm", dev = "pdf")
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.
```

```
xtabs( ~ rConf + RespType + StimType, recall)
```

Confidence – Recall

```
## , , StimType = Sham
##
##      RespType
## rConf Correct Foil Distractor
## 0         5     9         1
## 1       219   318       238
## 2       444   366       186
## 3       565   231        56
## 4      1026    90        18
##
## , , StimType = TI 1:3
##
##      RespType
```

```
## rConf Correct Foil Distractor
##      0      7      6      0
##      1     215  274     230
##      2     504  379     174
##      3     564  208      54
##      4    1091   79     12
```

```
# Removing confidence = 0 > no response
recall$rConf <- as.numeric(recall$rConf)
sum(recall$rConf=="0")
```

```
## [1] 28
```

```
Confidence_omission_Prct <- (sum(data_raw$rConf=="0")/length(data_raw$rConf))*100
Confidence_omission_Prct
```

```
## [1] 0.78125
```

```
recall <- recall%>% mutate(rConf = replace(rConf, rConf=="0", NA))
recall <- recall[!is.na(recall$rConf),]
```

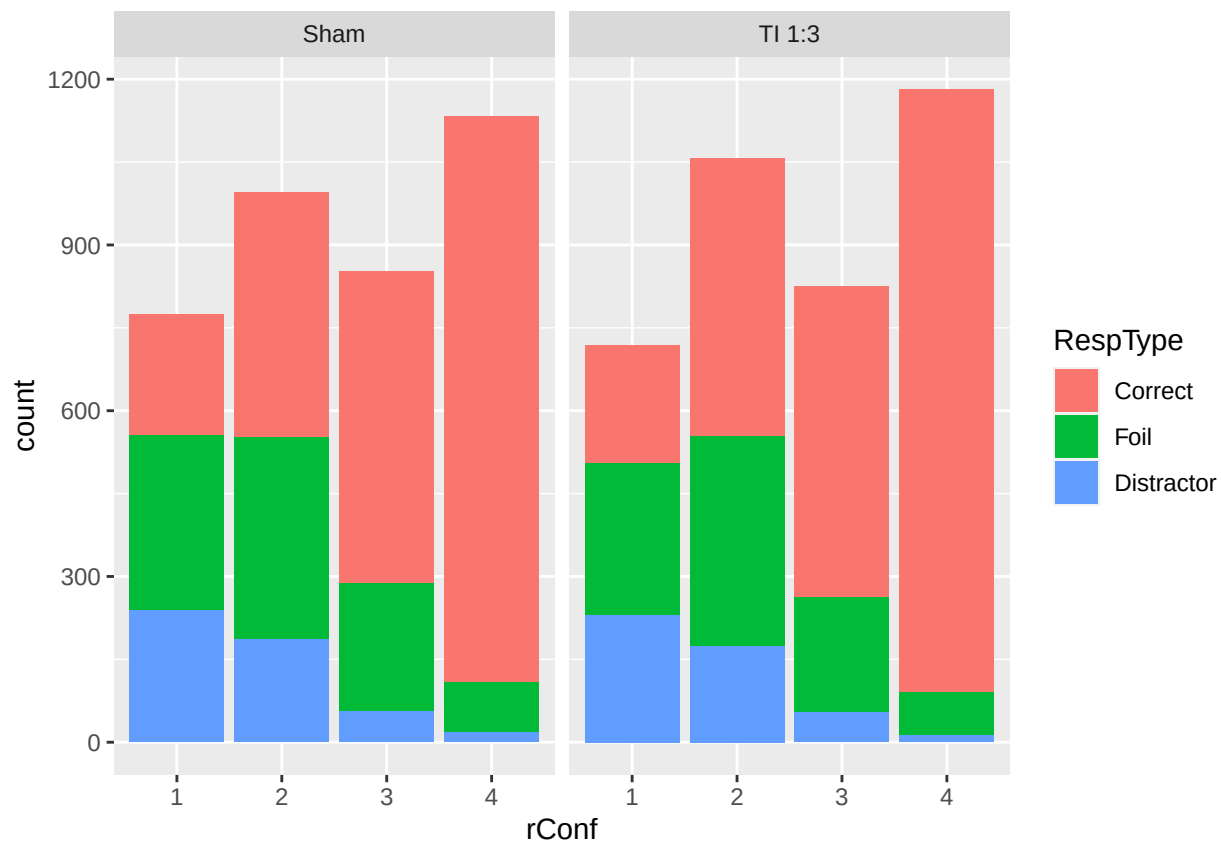
```
# Observed means
pander(recall %>%
  group_by(RespType, StimType) %>%
  summarise(mean(rConf), sd(rConf)))
```

```
## 'summarise()' has grouped output by 'RespType'. You can override using the
## '.groups' argument.
```

RespType	StimType	mean(rConf)	sd(rConf)
Correct	Sham	3.064	1.018
Correct	TI 1:3	3.066	1.015
Foil	Sham	2.093	0.947
Foil	TI 1:3	2.098	0.9167
Distractor	Sham	1.707	0.8064
Distractor	TI 1:3	1.677	0.7766

```
# Stats
recall$rConf <- as.factor(recall$rConf)

ggplot(data = recall) +
  geom_bar(mapping = aes(x = rConf, fill = RespType)) + facet_wrap(~StimType)
```



```
# Does the accuracy per response type vary for different stimulation conditions
library(ordinal)
```

```
##
## Attaching package: 'ordinal'
##
## The following objects are masked from 'package:brms':
##
##   ranef, VarCorr
##
## The following objects are masked from 'package:lme4':
##
##   ranef, VarCorr
##
## The following object is masked from 'package:dplyr':
##
##   slice
```

```
library(ggeffects)

recall$rConf <- factor(recall$rConf,
                       ordered=TRUE)

m1 = clmm(rConf ~ RespType*StimType + (1|S_ID), data = recall, Hess = TRUE)
```

```
m2 = clmm(rConf ~ RespType*StimType + (1|S_ID) + (1|block), data = recall, Hess = TRUE)
#m3 = clmm(rConf ~ RespType*StimType + (1|S_ID) + (1|session), data = recall, Hess = TRUE)
# cannot model session in random effects because only 2 levels are available

anova(m1,m2)
```

```
## Likelihood ratio tests of cumulative link models:
##
##      formula:                                link: threshold:
## m1 rConf ~ RespType * StimType + (1 | S_ID)      logit flexible
## m2 rConf ~ RespType * StimType + (1 | S_ID) + (1 | block) logit flexible
##
##      no.par   AIC  logLik LR.stat df Pr(>Chisq)
## m1          9 17754 -8868.2
## m2         10 17745 -8862.6  11.114  1  0.0008567 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(RVAideMemoire)
```

```
## *** Package RVAideMemoire v 0.9-81-2 ***
##
## Attaching package: 'RVAideMemoire'
##
## The following object is masked _by_ '.GlobalEnv':
##
##      se
##
## The following objects are masked from 'package:ppcor':
##
##      pcor, pcor.test
##
## The following object is masked from 'package:lme4':
##
##      dummy
```

```
Anova.clmm (m2)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: rConf
##              LR Chisq Df Pr(>Chisq)
## RespType      2006.10  2  <2e-16 ***
## StimType         0.35  1    0.5568
## RespType:StimType 0.42  2    0.8086
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m2)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
```

```
##
## formula: rConf ~ RespType * StimType + (1 | S_ID) + (1 | block)
## data: recall
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 7541 -8862.62 17745.24 1099(3297) 6.92e-03 2.8e+02
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.56098 0.7490
## block (Intercept) 0.01239 0.1113
## Number of groups: S_ID 21, block 12
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## RespTypeFoil -1.92750 0.07398 -26.053 <2e-16 ***
## RespTypeDistractor -2.56255 0.10000 -25.624 <2e-16 ***
## StimTypeTI 1:3 0.04454 0.05690 0.783 0.434
## RespTypeFoil:StimTypeTI 1:3 -0.02809 0.09980 -0.281 0.778
## RespTypeDistractor:StimTypeTI 1:3 -0.08682 0.13601 -0.638 0.523
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
## Estimate Std. Error z value
## 1|2 -2.6901 0.1753 -15.349
## 2|3 -0.9451 0.1725 -5.479
## 3|4 0.2838 0.1720 1.650
```

```
emmeans.conf <- emmeans(m2, ~ RespType | StimType, mode = "mean.class")
emmeans.conf
```

```
## StimType = Sham:
## RespType mean.class SE df asymp.LCL asymp.UCL
## Correct 3.09 0.0869 Inf 2.92 3.26
## Foil 2.05 0.0889 Inf 1.88 2.23
## Distractor 1.75 0.0824 Inf 1.59 1.91
##
## StimType = TI 1:3:
## RespType mean.class SE df asymp.LCL asymp.UCL
## Correct 3.11 0.0859 Inf 2.94 3.28
## Foil 2.06 0.0894 Inf 1.89 2.24
## Distractor 1.73 0.0821 Inf 1.57 1.89
##
## Confidence level used: 0.95
```

```
# Comparing confidence across correct and incorrect responses
#
```

```
m3 = clmm(rConf ~ as.factor(Accuracy)*StimType + (1|S_ID) + (1|block), data = recall, Hess = TRUE)
Anova.clmm(m3)
```

```
## Analysis of Deviance Table (Type II tests)
```

```
##
## Response: rConf
##
## LR Chisq Df Pr(>Chisq)
## as.factor(Accuracy) 1926.29 1 <2e-16 ***
## StimType 0.41 1 0.5205
## as.factor(Accuracy):StimType 0.24 1 0.6212
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m3)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: rConf ~ as.factor(Accuracy) * StimType + (1 | S_ID) + (1 | block)
## data: recall
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 7541 -8902.62 17821.23 742(2226) 2.18e-03 2.9e+02
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.57493 0.7582
## block (Intercept) 0.01388 0.1178
## Number of groups: S_ID 21, block 12
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## as.factor(Accuracy)1 2.110152 0.067500 31.262 <2e-16 ***
## StimTypeTI 1:3 0.002205 0.067948 0.032 0.974
## as.factor(Accuracy)1:StimTypeTI 1:3 0.043865 0.088775 0.494 0.621
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
## Estimate Std. Error z value
## 1|2 -0.5536 0.1764 -3.138
## 2|3 1.1704 0.1767 6.622
## 3|4 2.3957 0.1782 13.441
```

```
emmeans(m3, ~ StimType | Accuracy, mode = "mean.class")
```

```
## Accuracy = 0:
## StimType mean.class SE df asymp.LCL asymp.UCL
## Sham 1.96 0.0859 Inf 1.79 2.12
## TI 1:3 1.96 0.0861 Inf 1.79 2.13
##
## Accuracy = 1:
## StimType mean.class SE df asymp.LCL asymp.UCL
## Sham 3.08 0.0884 Inf 2.91 3.26
## TI 1:3 3.11 0.0873 Inf 2.94 3.28
##
## Confidence level used: 0.95
```



```
# Comparing confidence for correct responses only
recall_correct <- subset(recall, Accuracy=="1")
xtabs( ~ rConf + StimType, recall_correct)
```

```
##      StimType
## rConf Sham TI 1:3
##    1  219    215
##    2  444    504
##    3  565    564
##    4 1026   1091
```

```
m4 = clmm(rConf ~ StimType + (1|S_ID) + (1|block), data = recall, Hess = TRUE)
Anova.clmm(m4)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: rConf
##          LR Chisq Df Pr(>Chisq)
## StimType  3.3087  1   0.06891 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m4)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: rConf ~ StimType + (1 | S_ID) + (1 | block)
## data:    recall
##
## link threshold nobs logLik  AIC      niter    max.grad cond.H
## logit flexible 7541 -9865.88 19743.77 402(1203) 2.90e-03 3.8e+02
##
## Random effects:
## Groups Name      Variance Std.Dev.
## S_ID   (Intercept) 0.50488  0.7106
## block  (Intercept) 0.03868  0.1967
## Number of groups: S_ID 21, block 12
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## StimTypeTI 1:3  0.07648    0.04205   1.819   0.0689 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
##      Estimate Std. Error z value
## 1|2 -1.48684    0.16926  -8.784
## 2|3 -0.05033    0.16826  -0.299
## 3|4  0.96319    0.16858   5.714
```

```
emmeans(m4, ~ StimType)
```

```
## StimType emmean SE df asymp.LCL asymp.UCL
## Sham 0.191 0.168 Inf -0.1376 0.520
## TI 1:3 0.268 0.168 Inf -0.0611 0.597
##
## Confidence level used: 0.95
```

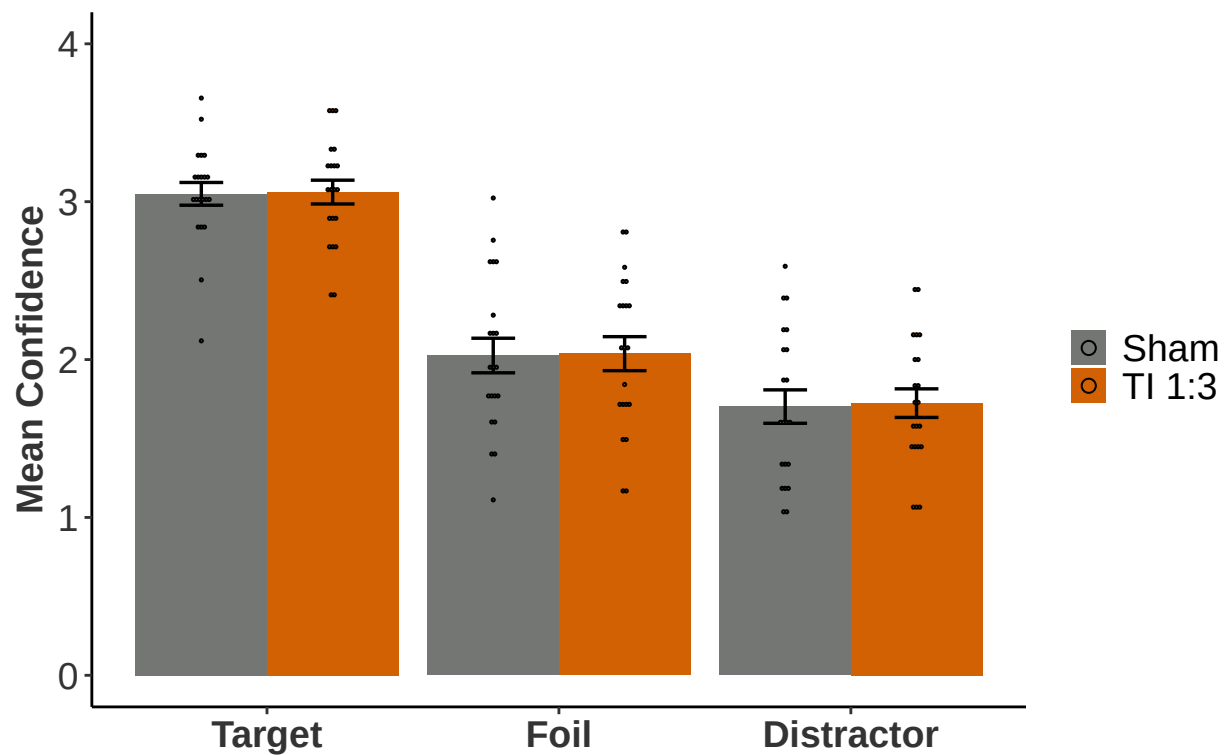
```
## Plot Confidence
```

```
g_data <- ddply(recall, c("S_ID", "StimType", "RespType"), summarise,
  sum_acc = sum(Accuracy),
  sum_acc_perc = (sum_acc/192)*100,
  mean_acc = mean(Accuracy),
  mean_acc_perc = mean_acc*100,
  sd_acc = sd(Accuracy),
  se_acc = se(Accuracy),
  mean_conf = mean(Confidence)
)
```

```
g_data$RespType <- factor(g_data$RespType, levels = c("Correct", "Foil", "Distractor"), ordered = TRUE)
```

```
p.Conf <- ggplot(g_data, aes(x = RespType, y = mean_conf, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.2) +
  scale_y_continuous(name = "Mean Confidence",
    breaks = seq(0, 4, 1),
    limits=c(0, 4)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Correct","Foil", "Distractor"),
    labels=c("Target","Foil", "Distractor")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size=14),
    legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("Sham" = "#f08080", "TI 1:3" = "#4682b4"))
p.Conf
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.
```



```
ggsave(p.Conf, filename = "Figure_5F.pdf", width = 11, height = 9, dpi = 300, units = "cm", device = "pdf")
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.
```

```
retest <- subset(bdata, Run == "2")
xtabs( ~ RespType + StimType, retest)
```

Accuracy - Re-test

```
##           StimType
## RespType  Sham TI 1:3
##   Correct   1793  1883
##   Foil      1046  1030
##   Distractor  922   883
```

```
# ----- Multinomial logistic regression -----
# Does the number of responses per response type vary for different stimulation conditions
retest$RespType <- relevel(retest$RespType, ref = "Correct")
m.acc <- nnet::multinom(RespType ~ StimType, data = retest)
```

```
## # weights: 9 (4 variable)
## initial value 8302.213065
## iter 10 value 7914.463459
## iter 10 value 7914.463454
## final value 7914.463454
## converged
```

```
summary(m.acc)
```

```
## Call:
## nnet::multinom(formula = RespType ~ StimType, data = retest)
##
## Coefficients:
##          (Intercept) StimTypeTI 1:3
## Foil          -0.5389910      -0.06442160
## Distractor    -0.6652348      -0.09216158
##
## Std. Errors:
##          (Intercept) StimTypeTI 1:3
## Foil           0.03890678      0.05491541
## Distractor     0.04052630      0.05749764
##
## Residual Deviance: 15828.93
## AIC: 15836.93
```

```
exp(coef(m.acc))
```

```
##          (Intercept) StimTypeTI 1:3
## Foil           0.5833365      0.9376096
## Distractor     0.5141528      0.9119578
```

```
# Check the Z-score for the model (wald Z)
z <- summary(m.acc)$coefficients/summary(m.acc)$standard.errors
z
```

```
##          (Intercept) StimTypeTI 1:3
## Foil          -13.85340      -1.173106
## Distractor    -16.41489      -1.602876
```

```
p <- (1 - pnorm(abs(z), 0, 1))*2
p
```

```
##          (Intercept) StimTypeTI 1:3
## Foil              0      0.2407533
## Distractor        0      0.1089621
```

```
Anova(m.acc)
```

```
## # weights: 6 (2 variable)
## initial value 8302.213065
## final value 7915.967264
## converged
```

```

## Analysis of Deviance Table (Type II tests)
##
## Response: RespType
##           LR Chisq Df Pr(>Chisq)
## StimType   3.0076  2    0.2223

#----- Binomial -----
m1 <- glmer(Accuracy ~ StimType + (1|S_ID) + (1|session), data=retest, family = binomial(link = "logit"))
m2<- glmer(Accuracy ~ StimType + (1|S_ID) + (1|session) + (1|block), data=retest, family = binomial(link = "logit"))
m3 <- glmer(Accuracy ~ StimType + (1|S_ID) + (1|session) + (1|block/trial), data=retest, family = binomial(link = "logit"))
anova(m1, m2, m3)

## Data: retest
## Models:
## m1: Accuracy ~ StimType + (1 | S_ID) + (1 | session)
## m2: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block)
## m3: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##      npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## m1      4 9962.3  9990 -4977.1   9954.3
## m2      5 9959.4  9994 -4974.7   9949.4 4.9192  1    0.02656 *
## m3      6 9961.4 10003 -4974.7   9949.4 0.0019  1    0.96511
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

lmtest::lrtest(m1, m2, m3)

## Likelihood ratio test
##
## Model 1: Accuracy ~ StimType + (1 | S_ID) + (1 | session)
## Model 2: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block)
## Model 3: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##      #Df LogLik Df  Chisq Pr(>Chisq)
## 1      4 -4977.1
## 2      5 -4974.7  1 4.9192    0.02656 *
## 3      6 -4974.7  1 0.0019    0.96511
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# keeping m2
Anova(m2)

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: Accuracy
##           Chisq Df Pr(>Chisq)
## StimType 1.8392  1    0.175

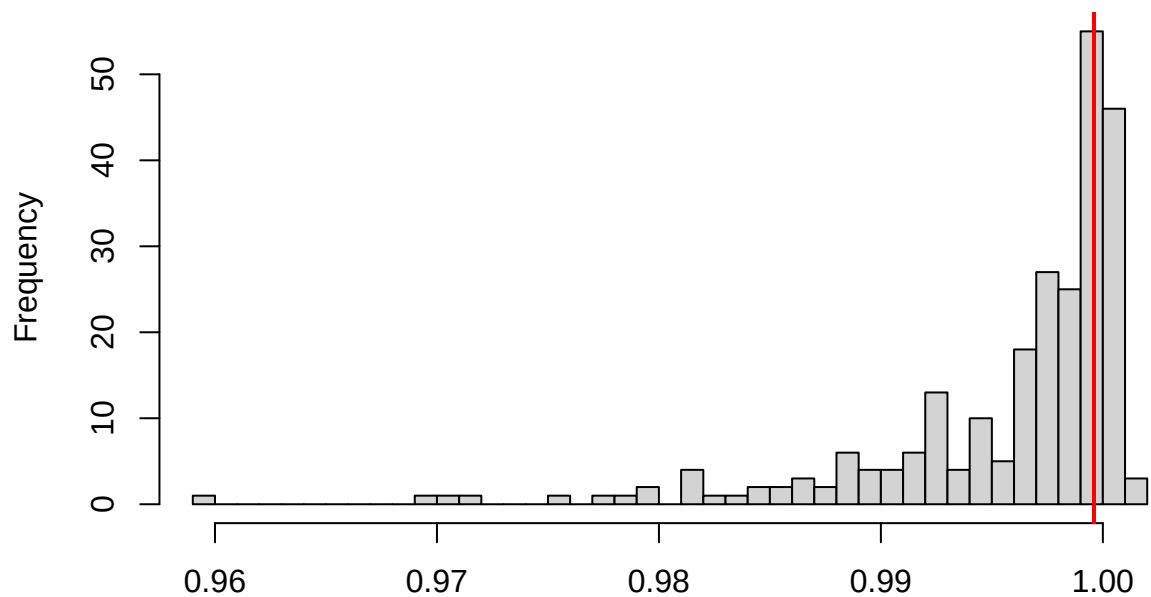
```

```
summary(m2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##   Family: binomial   ( logit )
## Formula: Accuracy ~ StimType + (1 | S_ID) + (1 | session) + (1 | block)
##   Data: retest
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##           AIC          BIC    logLik deviance df.resid
##    9959.4    9994.0   -4974.7   9949.4     7552
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.7719 -0.9103 -0.5498  0.9808  1.9662
##
## Random effects:
##   Groups Name            Variance Std.Dev.
##   S_ID   (Intercept) 0.339143 0.58236
##   block  (Intercept) 0.008674 0.09313
##   session (Intercept) 0.003256 0.05706
## Number of obs: 7557, groups: S_ID, 21; block, 12; session, 2
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.08496    0.14025  -0.606    0.545
## StimTypeTI 1:3  0.06515    0.04804   1.356    0.175
##
## Correlation of Fixed Effects:
##              (Intr)
## StmTypTI1:3 -0.173
```

```
library(DHARMA)
testDispersion(m2)
```

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

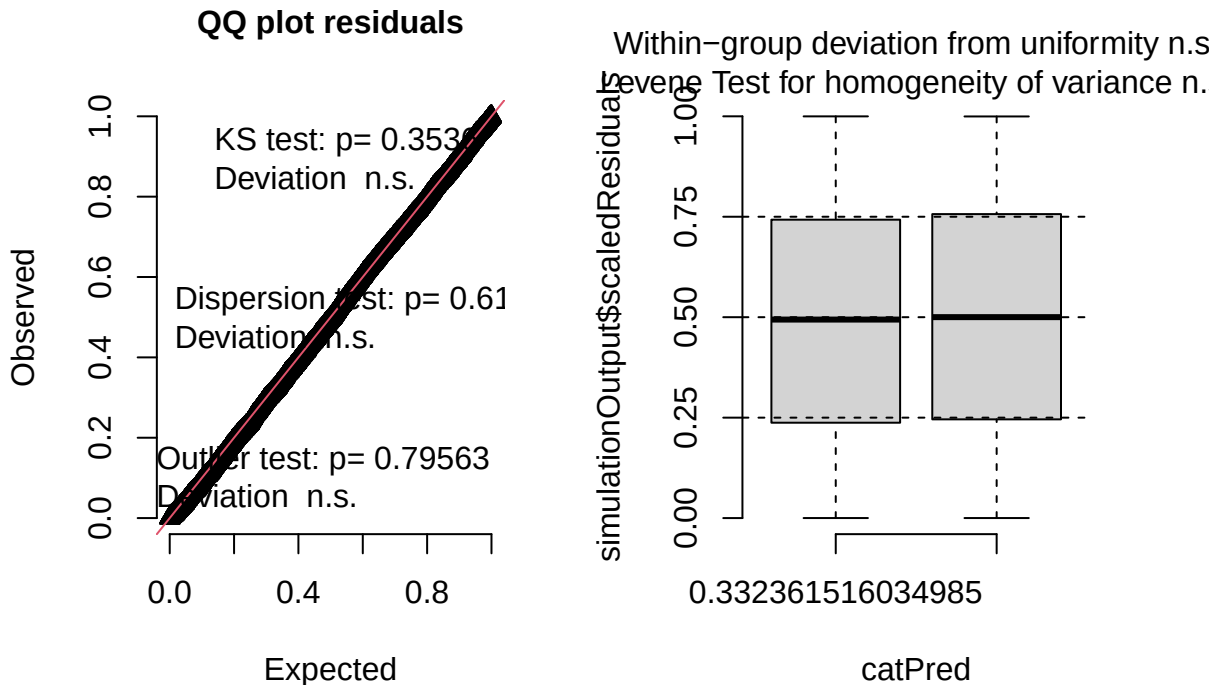


Simulated values, red line = fitted model. p-value (two.sided) = 0.616

```
##  
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.  
## simulated  
##  
## data: simulationOutput  
## dispersion = 1.0038, p-value = 0.616  
## alternative hypothesis: two.sided
```

```
simulationOutput <- simulateResiduals(fittedModel = m2, plot = T)
```

DHARMA residual



Plot - Fig S8

```
g_bdata <- ddply(retest, c("S_ID", "StimType", "RespType"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/192,
  meanRespTypePerc = meanRespType*100,
  medianRT = median(RT, na.rm=TRUE),
  meanRT = mean(RT, na.rm=TRUE),
  mean_Conf = mean(Confidence, na.rm=TRUE)
)
```

```
p.RespType <- ggplot(g_bdata, aes(x = RespType, y = meanRespTypePerc, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "Mean Response (%)",
    breaks = seq(0, 100, 20),
    limits=c(0, 100)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
```



```

    legend.title=element_blank() +
    scale_x_discrete(breaks=c("Correct","Foil", "Distractor"),
    labels=c("Target","Foil", "Distractor")) +
    theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size=14),
    legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("

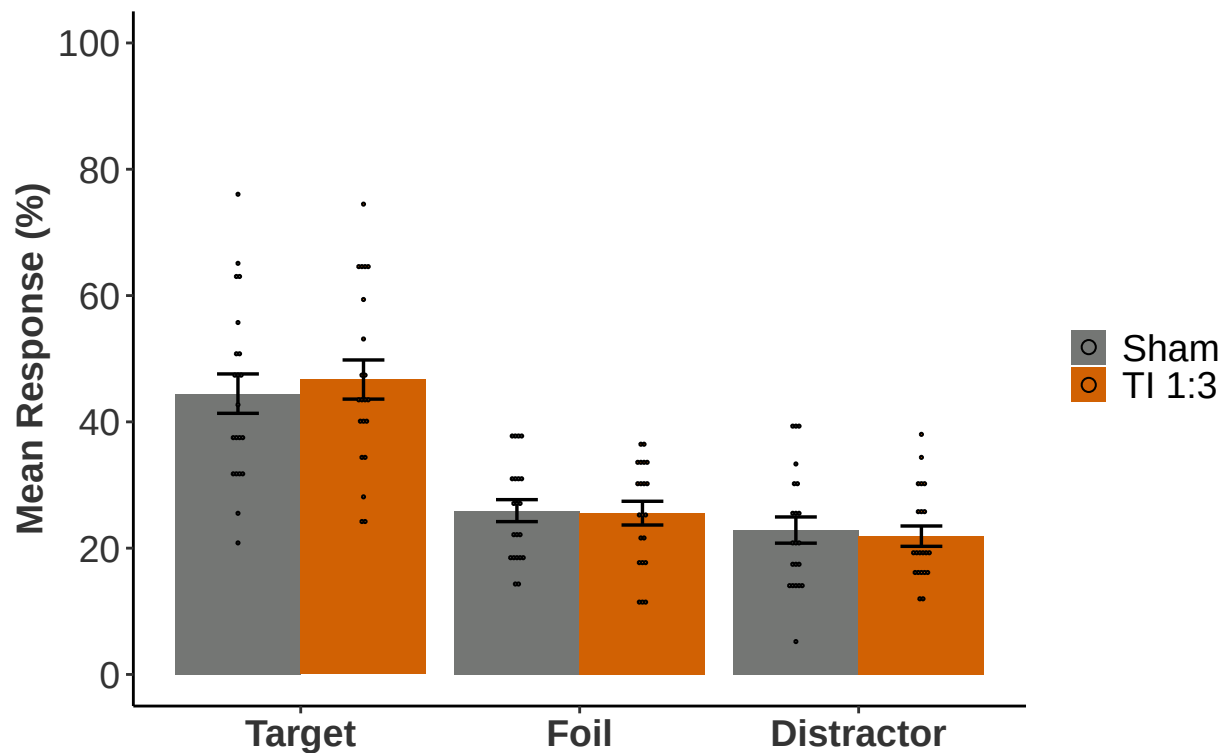
```

p.RespType

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

```



```

ggsave(p.RespType, filename = "Figure_S8A.tiff", width = 11, height = 9, dpi = 300, units = "cm", devi

```

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

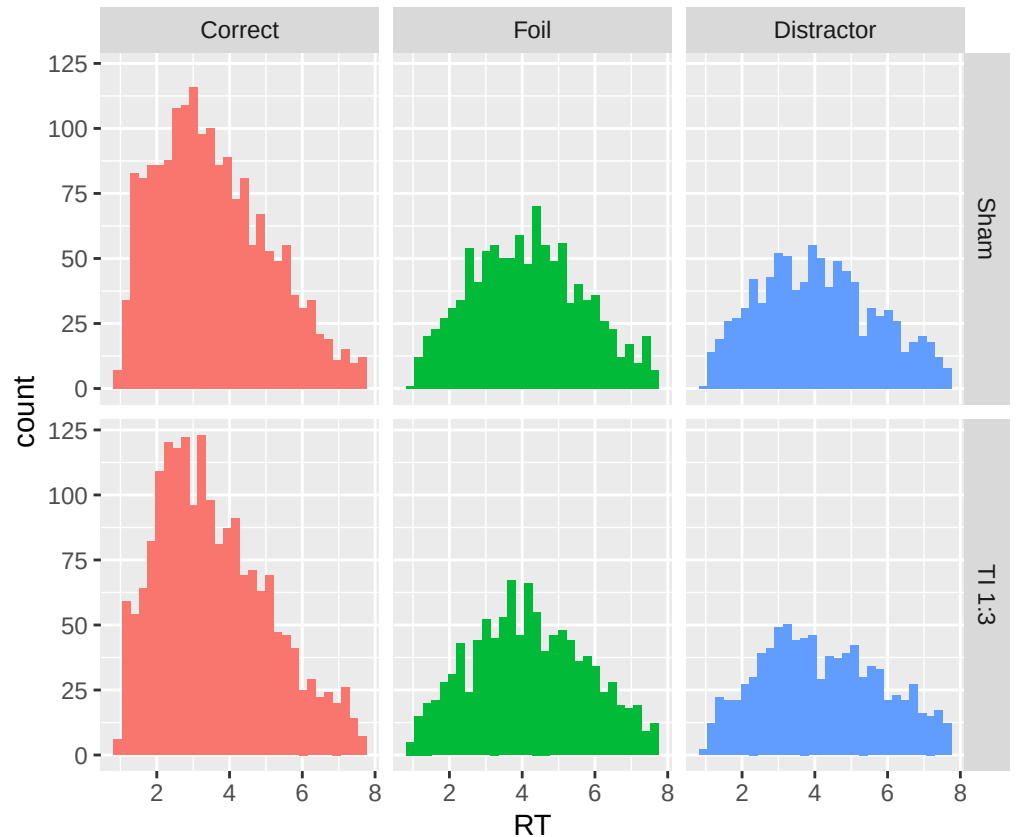
```

```

# plotting the data
library(ggribes)

```

```
ggplot(retest, aes(x = RT, fill = RespType)) +
  geom_histogram(bins = 30) +
  facet_grid(StimType ~ RespType)
```



Reaction Times - Re-test

```
pander(retest %>%
  group_by(RespType, StimType) %>%
  summarise(median(RT), sd(RT)))
```

'summarise()' has grouped output by 'RespType'. You can override using the
'.groups' argument.

RespType	StimType	median(RT)	sd(RT)
Correct	Sham	3.358	1.523
Correct	TI 1:3	3.318	1.537
Foil	Sham	4.113	1.531
Foil	TI 1:3	4.119	1.572
Distractor	Sham	3.997	1.599
Distractor	TI 1:3	4.012	1.643

Does reaction time per response type vary for different stimulation conditions?

Using glm without random effects to find starting values for the optimisation

```

m0 <- glm(RT ~ StimType*RespType, data=retest, family = inverse.gaussian(link=identity))

m1 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|session),
  data = retest,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m2 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|block) + (1|session),
  data = retest,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m3 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|session) + (1|block/trial),
  data = retest,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

anova(m1, m2, m3)

```

```

## Data: retest
## Models:
## m1: RT ~ StimType * RespType + (1 | S_ID) + (1 | session)
## m2: RT ~ StimType * RespType + (1 | S_ID) + (1 | block) + (1 | session)
## m3: RT ~ StimType * RespType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##      npar   AIC    BIC logLik deviance  Chisq Df Pr(>Chisq)
## m1      9 26606 26668 -13294    26588
## m2     10 26588 26657 -13284    26568 20.211  1  6.934e-06 ***
## m3     11 26560 26636 -13269    26538 29.630  1  5.229e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

lmtest::lrtest(m1, m2, m3)

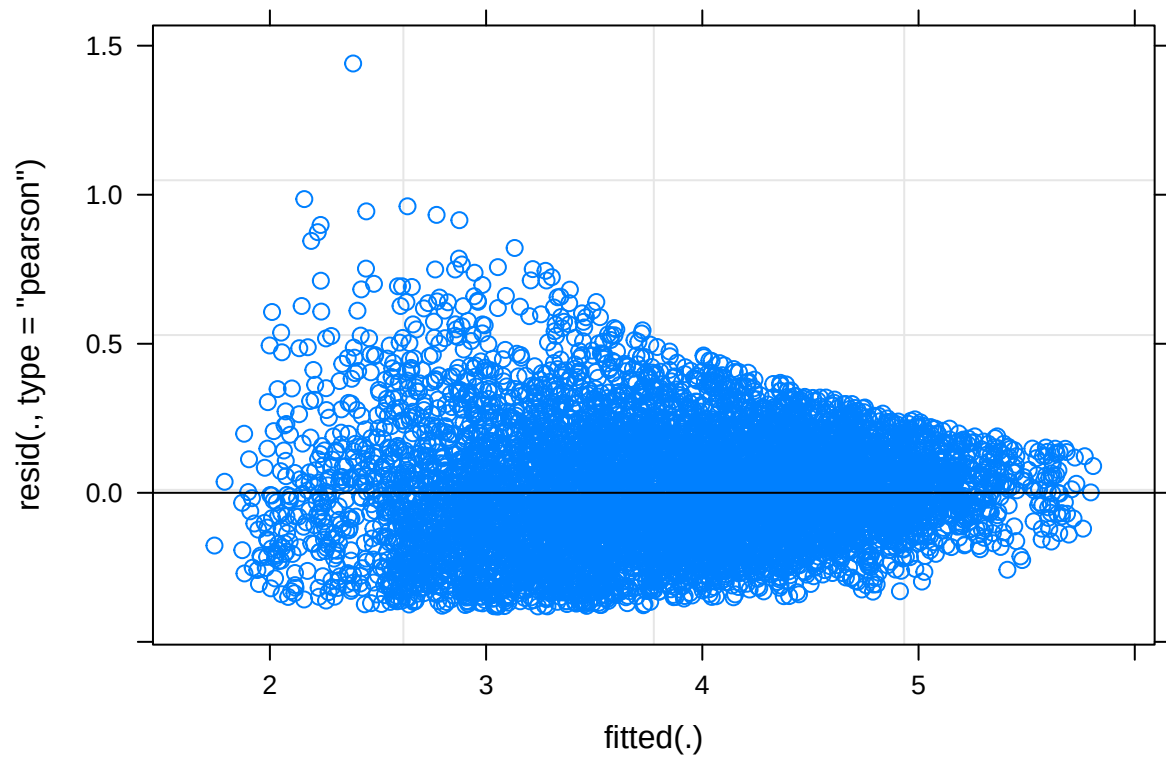
```

```

## Likelihood ratio test
##
## Model 1: RT ~ StimType * RespType + (1 | S_ID) + (1 | session)
## Model 2: RT ~ StimType * RespType + (1 | S_ID) + (1 | block) + (1 | session)
## Model 3: RT ~ StimType * RespType + (1 | S_ID) + (1 | session) + (1 |
##      block/trial)
##      #Df LogLik Df  Chisq Pr(>Chisq)
## 1      9 -13294
## 2     10 -13284  1 20.211  6.934e-06 ***
## 3     11 -13269  1 29.630  5.229e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

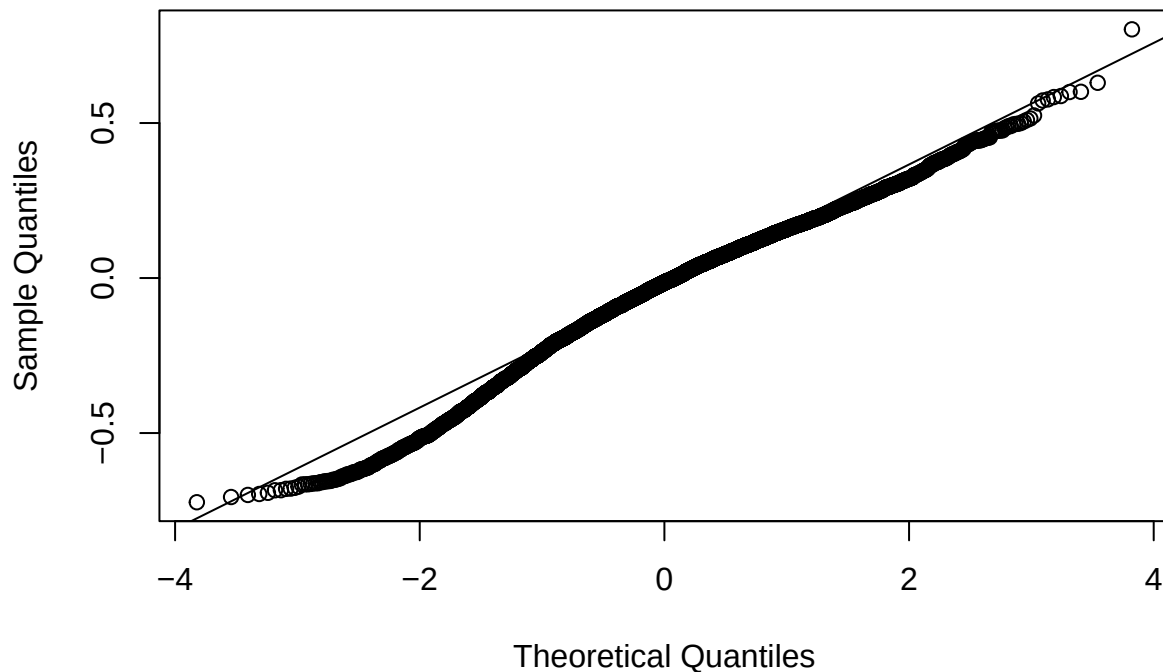
```

```
# keeping m3  
plot(m3)
```



```
qqnorm(resid(m3))  
qqline(resid(m3))
```

Normal Q-Q Plot



```
Anova(m3)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##               Chisq Df Pr(>Chisq)
## StimType       0.2437  1    0.6215
## RespType      291.3012  2   <2e-16 ***
## StimType:RespType 0.3386  2    0.8443
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# -- Does reaction time for correct vs incorrect vary for different stimulation conditions
m0 <- glm(RT ~ StimType*Accuracy, data=retest, family = inverse.gaussian(link=identity))

m2 <- glmer(RT ~ StimType*Accuracy + (1|S_ID) + (1|session) + (1|block),
  data = retest,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m3 <- glmer(RT ~ StimType*Accuracy + (1|S_ID) + (1|session) + (1|block/trial),
  data = retest,
  family = inverse.gaussian(link=identity),
```

```

start=list(fixef=coef(m0)),
control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

anova(m0, m2, m3)

## Analysis of Deviance Table
##
## Model: inverse.gaussian, link: identity
##
## Response: RT
##
## Terms added sequentially (first to last)
##
##
##           Df Deviance Resid. Df Resid. Dev
## NULL                        7556      436.05
## StimType          1   0.0274      7555      436.02
## Accuracy          1  10.8541      7554      425.17
## StimType:Accuracy  1   0.0166      7553      425.15

lmtest::lrtest(m0, m2, m3)

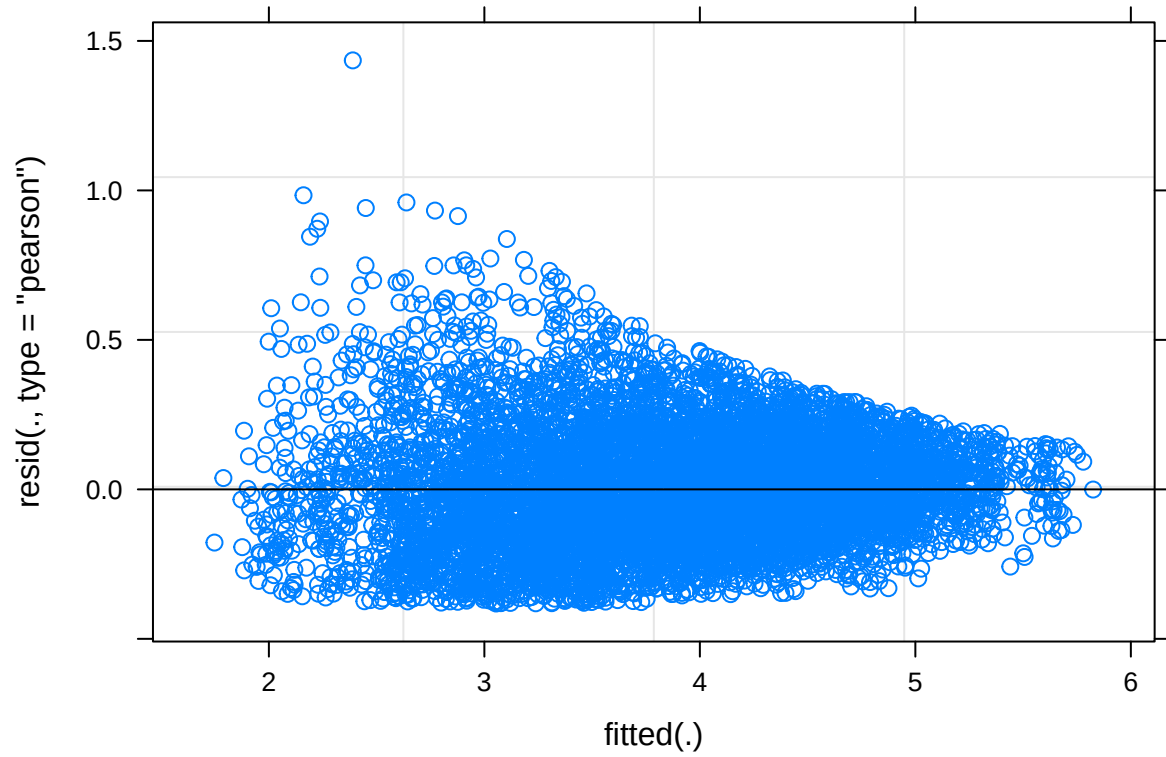
## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"

## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"

## Likelihood ratio test
##
## Model 1: RT ~ StimType * Accuracy
## Model 2: RT ~ StimType * Accuracy + (1 | S_ID) + (1 | session) + (1 |
##           block)
## Model 3: RT ~ StimType * Accuracy + (1 | S_ID) + (1 | session) + (1 |
##           block/trial)
##   #Df LogLik Df    Chisq Pr(>Chisq)
## 1    5 -14133
## 2    8 -13285  3 1695.993 < 2.2e-16 ***
## 3    9 -13270  1   29.637 5.211e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

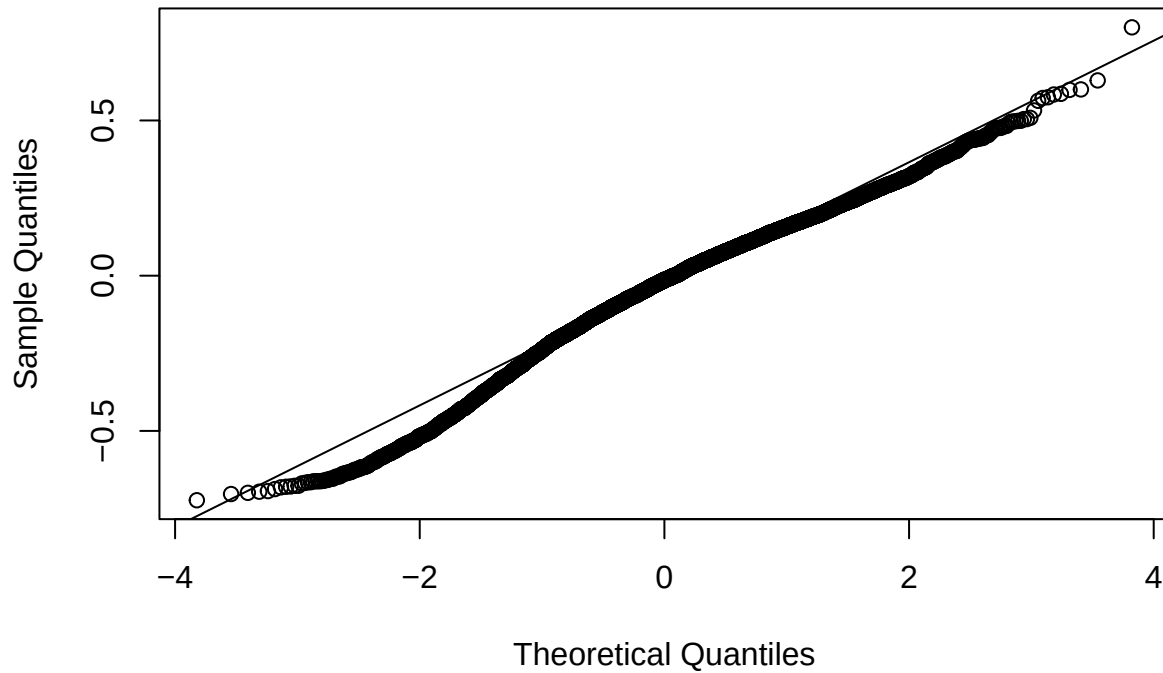
plot(m3)

```



```
qqnorm(resid(m3))  
qqline(resid(m3))
```

Normal Q-Q Plot



```
Anova(m3)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##               Chisq Df Pr(>Chisq)
## StimType       0.2549  1    0.6136
## Accuracy      290.5765  1   <2e-16 ***
## StimType:Accuracy  0.3373  1    0.5614
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

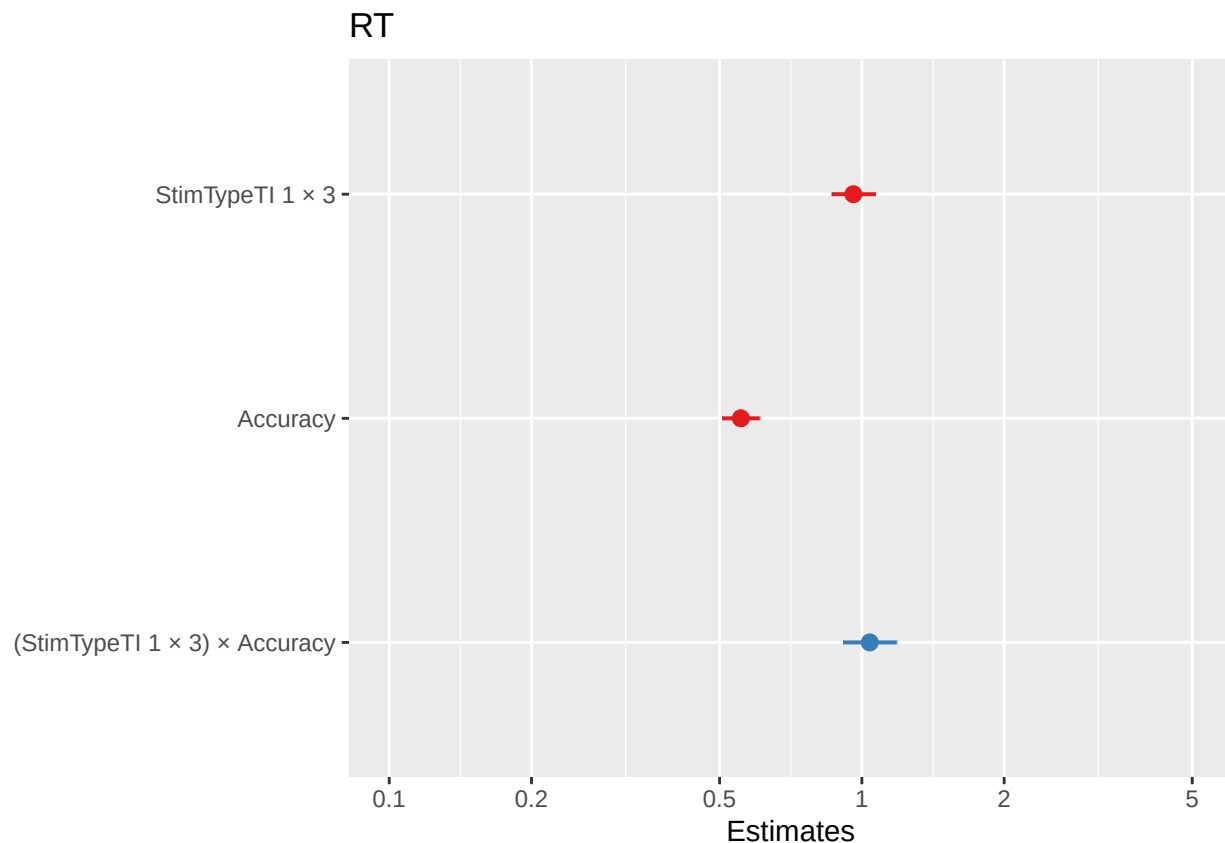
```
summary(m3)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
## Family: inverse.gaussian ( identity )
## Formula: RT ~ StimType * Accuracy + (1 | S_ID) + (1 | session) + (1 |
##   block/trial)
## Data: retest
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
##   nAGQ0initStep = FALSE)
##
##      AIC      BIC    logLik deviance df.resid
## 26557.9 26620.2 -13269.9 26539.9      7548
```



```
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.9529 -0.6928 -0.0725  0.6045  7.3397
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## trial:block (Intercept) 0.018765 0.13699
## S_ID        (Intercept) 0.082573 0.28736
## block       (Intercept) 0.003084 0.05554
## session     (Intercept) 0.003727 0.06105
## Residual                    0.038223 0.19551
## Number of obs: 7557, groups: trial:block, 192; S_ID, 21; block, 12; session, 2
##
## Fixed effects:
##              Estimate Std. Error t value Pr(>|z|)
## (Intercept)      4.55146    0.29768  15.290  <2e-16 ***
## StimTypeTI 1:3      -0.04140    0.05383   -0.769    0.442
## Accuracy           -0.58960    0.04616  -12.773  <2e-16 ***
## StimTypeTI 1:3:Accuracy  0.03850    0.06630    0.581    0.561
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) StTTI1:3 Accrcy
## StmTypTI1:3 -0.105
## Accuracy     -0.092  0.515
## StmTTI1:3:A  0.063 -0.733  -0.688
```

```
plot_model(m3)
```



```
estimated_means <- emmeans(m3, specs = "StimType")
```

NOTE: Results may be misleading due to involvement in interactions

```
estimated_means
```

```
## StimType emmean SE df asymp.LCL asymp.UCL
## Sham 4.26 0.296 Inf 3.68 4.84
## TI 1:3 4.23 0.296 Inf 3.66 4.81
##
## Results are averaged over the levels of: Accuracy
## Confidence level used: 0.95
```

```
# Check RT for correct Trials only -
```

```
#
```

```
retest_correct <- subset(retest, Accuracy=="1")
```

```
m0 <- glm(RT ~ StimType, data=retest_correct, family = inverse.gaussian(link=identity))
```

```
m2 <- glmer(RT ~ StimType + (1|S_ID) + (1|block) + (1|session),
  data = retest_correct,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))
```

```

m3 <- glmer(RT ~ StimType + (1|S_ID) + (1|session) + (1|block/trial),
  data = retest_correct,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

```

```
anova(m0, m2, m3)
```

```

## Analysis of Deviance Table
##
## Model: inverse.gaussian, link: identity
##
## Response: RT
##
## Terms added sequentially (first to last)
##
##
##          Df  Deviance Resid. Df Resid. Dev
## NULL                        3675      234.17
## StimType  1 0.0056795      3674      234.16

```

```
lmtest::lrtest(m0, m2, m3)
```

```

## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"

```

```

## Warning in modelUpdate(objects[[i - 1]], objects[[i]]): original model was of
## class "glm", updated model is of class "glmerMod"

```

```

## Likelihood ratio test
##
## Model 1: RT ~ StimType
## Model 2: RT ~ StimType + (1 | S_ID) + (1 | block) + (1 | session)
## Model 3: RT ~ StimType + (1 | S_ID) + (1 | session) + (1 | block/trial)
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1    3 -6642.9
## 2    6 -6322.4  3 640.924 < 2.2e-16 ***
## 3    7 -6313.9  1  17.139  3.475e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
Anova(m2)
```

```

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##           Chisq Df Pr(>Chisq)
## StimType 9e-04  1    0.9758

```

```
summary(m2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: inverse.gaussian ( identity )
## Formula: RT ~ StimType + (1 | S_ID) + (1 | block) + (1 | session)
## Data: retest_correct
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
## nAGQ0initStep = FALSE)
##
##      AIC      BIC    logLik deviance df.resid
## 12656.9 12694.1 -6322.4 12644.9     3670
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.7729 -0.7210 -0.1283  0.6170  6.3415
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## S_ID      (Intercept)  0.102190  0.31967
## block     (Intercept)  0.005847  0.07647
## session   (Intercept)  0.006663  0.08163
## Residual                    0.046372  0.21534
## Number of obs: 3676, groups: S_ID, 21; block, 12; session, 2
##
## Fixed effects:
##              Estimate Std. Error t value Pr(>|z|)
## (Intercept)    3.87087    0.31177   12.42  <2e-16 ***
## StimTypeTI 1:3 -0.00152    0.05009   -0.03    0.976
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## StmTypTI1:3 -0.105
```

```
## Plot RT re-test
```

```
##
```

```
g_bdata <- ddply(retest, c("S_ID", "StimType", "RespType"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/192,
  meanRespTypePerc = meanRespType*100,
  medianRT = median(RT, na.rm=TRUE),
  meanRT = mean(RT, na.rm=TRUE),
  mean_Conf = mean(Confidence, na.rm=TRUE)
)
```

```
p.RespTypeRT <- ggplot(g_bdata, aes(x = RespType, y = medianRT, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "Median RT (s)",
    breaks = seq(0, 8, 2),
    limits=c(0, 8)) +
```

```

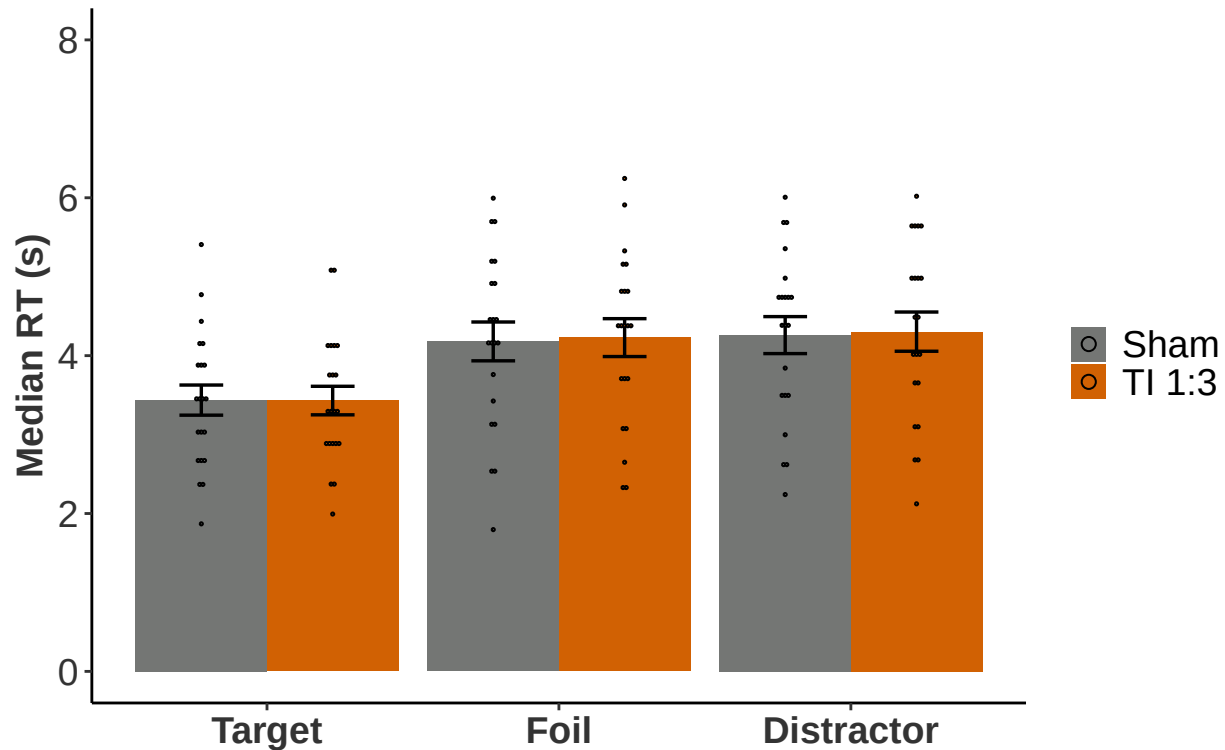
theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
      axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
      axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
      axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
      axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
      legend.text = element_blank(),
      legend.background = element_blank(),
      legend.key=element_blank(),
      panel.background = element_blank(),
      legend.title=element_blank()) +
scale_x_discrete(breaks=c("Correct","Foil", "Distractor"),
  labels=c("Target","Foil", "Distractor")) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 20),
        legend.text = element_text(size=14),
        legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("
p.RespTypeRT

```

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

```



```
ggsave(p.RespTypeRT, filename = "Figure_S8B.tiff", width = 11, height = 9, dpi = 300, units = "cm", de
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.
```

```
g_bdata <- ddply(bdata, c("S_ID", "StimType", "RespType", "Run"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/192,
  meanRespTypePerc = meanRespType*100,
)

### Means to report
g_bdata %>%
  group_by(StimType, RespType, Run) %>%
  summarise_at(vars(meanRespType), list(mean = mean, std = sd))
```

Recall and Re-test

```
## # A tibble: 12 x 5
## # Groups:   StimType, RespType [6]
##   StimType RespType Run    mean    std
##   <fct>    <fct>   <fct> <dbl> <dbl>
## 1 Sham     Correct    1     0.560 0.150
## 2 Sham     Correct    2     0.445 0.143
## 3 Sham     Foil       1     0.251 0.0898
## 4 Sham     Foil       2     0.259 0.0797
## 5 Sham     Distractor 1     0.124 0.0848
## 6 Sham     Distractor 2     0.229 0.0951
## 7 TI 1:3   Correct    1     0.591 0.144
## 8 TI 1:3   Correct    2     0.467 0.142
## 9 TI 1:3   Foil       1     0.235 0.100
## 10 TI 1:3  Foil       2     0.255 0.0861
## 11 TI 1:3  Distractor 1     0.117 0.0702
## 12 TI 1:3  Distractor 2     0.219 0.0738
```

```
bdata$RespType <- relevel(bdata$RespType, ref = "Correct")
m.acc <- nnet::multinom(RespType ~ StimType*Run, data = bdata)
```

```
## # weights: 15 (8 variable)
## initial value 16617.609478
## iter 10 value 14824.419147
## final value 14821.872844
## converged
```

```
summary(m.acc)
```

```
## Call:
## nnet::multinom(formula = RespType ~ StimType * Run, data = bdata)
```

```
##
## Coefficients:
##      (Intercept) StimTypeTI 1:3      Run2 StimTypeTI 1:3:Run2
## Foil      -0.8010328    -0.1220075 0.2621147      0.05760720
## Distractor -1.5100896    -0.1124642 0.8449852      0.02026912
##
## Std. Errors:
##      (Intercept) StimTypeTI 1:3      Run2 StimTypeTI 1:3:Run2
## Foil      0.03780042    0.05390685 0.05424591      0.07695220
## Distractor 0.04946416    0.07067085 0.06394551      0.09110577
##
## Residual Deviance: 29643.75
## AIC: 29659.75
```

```
exp(coef(m.acc))
```

```
##      (Intercept) StimTypeTI 1:3      Run2 StimTypeTI 1:3:Run2
## Foil      0.4488651    0.8851417 1.299676      1.059299
## Distractor 0.2208902    0.8936294 2.327943      1.020476
```

```
# Check the Z-score for the model (wald Z)
z <- summary(m.acc)$coefficients/summary(m.acc)$standard.errors
z
```

```
##      (Intercept) StimTypeTI 1:3      Run2 StimTypeTI 1:3:Run2
## Foil      -21.19111    -2.263303 4.831971      0.7486101
## Distractor -30.52896    -1.591380 13.214143      0.2224790
```

```
p <- (1 - pnorm(abs(z), 0, 1))*2
p
```

```
##      (Intercept) StimTypeTI 1:3      Run2 StimTypeTI 1:3:Run2
## Foil      0      0.02361704 1.351878e-06      0.4540922
## Distractor 0      0.11152417 0.000000e+00      0.8239410
```

```
Anova(m.acc)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: RespType
##      LR Chisq Df Pr(>Chisq)
## StimType      8.80 2    0.01228 *
## Run      371.45 2    < 2e-16 ***
## StimType:Run    0.56 2    0.75542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

# Check answers face-name pair recall to re-test
NameC1 <- ifelse(recall$Accuracy==1, recall$Name, NA)
recall$NameC1 <- NameC1

id = unique(bdata$S_ID)
cond <- vector()
for (i in 1:length(unique(retest$S_ID))) {
  for (s in 1:length(unique(retest$session))) {
    tmp_df = subset(recall, S_ID==id[i] & session==s)
    tmp_df2 = subset(retest, S_ID==id[i] & session==s)
    print(paste("Subject", id[i], "session", s))
    ntrial_tmp = length(unique(tmp_df2$Name))
    for (n in 1:ntrial_tmp) {
      m <- match(tmp_df2$Name[n], tmp_df$NameC1)
      if (!is.na(m)){
        cond <- c(cond, 1)
      } else {
        cond <- c(cond, 0)
      }
    }
  }
}

```

Recall - Re-test detailed

```

## [1] "Subject 1 session 1"
## [1] "Subject 1 session 2"
## [1] "Subject 2 session 1"
## [1] "Subject 2 session 2"
## [1] "Subject 3 session 1"
## [1] "Subject 3 session 2"
## [1] "Subject 4 session 1"
## [1] "Subject 4 session 2"
## [1] "Subject 5 session 1"
## [1] "Subject 5 session 2"
## [1] "Subject 6 session 1"
## [1] "Subject 6 session 2"
## [1] "Subject 7 session 1"
## [1] "Subject 7 session 2"
## [1] "Subject 8 session 1"
## [1] "Subject 8 session 2"
## [1] "Subject 9 session 1"
## [1] "Subject 9 session 2"
## [1] "Subject 10 session 1"
## [1] "Subject 10 session 2"
## [1] "Subject 11 session 1"
## [1] "Subject 11 session 2"
## [1] "Subject 12 session 1"
## [1] "Subject 12 session 2"
## [1] "Subject 13 session 1"
## [1] "Subject 13 session 2"
## [1] "Subject 14 session 1"
## [1] "Subject 14 session 2"
## [1] "Subject 15 session 1"

```



```
## [1] "Subject 15 session 2"
## [1] "Subject 16 session 1"
## [1] "Subject 16 session 2"
## [1] "Subject 17 session 1"
## [1] "Subject 17 session 2"
## [1] "Subject 18 session 1"
## [1] "Subject 18 session 2"
## [1] "Subject 19 session 1"
## [1] "Subject 19 session 2"
## [1] "Subject 20 session 1"
## [1] "Subject 20 session 2"
## [1] "Subject 21 session 1"
## [1] "Subject 21 session 2"
```

```
retest$Correct_Rec <- cond
```

```
### Re-test memory fates
```

```
retest$type_memory = ifelse(retest$Accuracy==1 & retest$Correct_Rec==1,"Remembered",
                             ifelse(retest$Accuracy==0 & retest$Correct_Rec==1, "Forgotten",
                                     ifelse(retest$Accuracy==1 & retest$Correct_Rec==0, "Retrieved",
                                             ifelse(retest$Accuracy==0 & retest$Correct_Rec==0, "Never",NA)))
```

```
retest$type_transition = ifelse(retest$Accuracy==1 & retest$Correct_Rec==1,"T2T",
                                ifelse(retest$RespType=="Foil" & retest$Correct_Rec==1, "T2F",
                                        ifelse(retest$RespType=="Distractor" & retest$Correct_Rec==1, "T2D",
                                                ifelse(retest$Accuracy==1 & retest$Correct_Rec==0, "CR",
                                                        ifelse(retest$Accuracy==0 & retest$Correct_Rec==0, "Never",NA))))
```

```
retest$remembered = ifelse(retest$Accuracy==1 & retest$Correct_Rec==1,1,
                            ifelse(retest$Accuracy==0 & retest$Correct_Rec==1, 0,
                                    ifelse(retest$Accuracy==1 & retest$Correct_Rec==0, 0,
                                            ifelse(retest$Accuracy==0 & retest$Correct_Rec==0, 0,NA))))
```

```
## What happens at re-test to the face-name associations previously learned (i.e. target at recall)?
```

```
knew <- subset(retest, Correct_Rec=="1")
knew.g <- ddpby(knew, c("S_ID","StimType"), summarise,
               t1 = sum(Correct_Rec),
               t2 = sum(remembered)
)
```

```
library(data.table)
```

```
##
```

```
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:reshape2':
```

```
##
```

```
## dcast, melt
```

```
## The following objects are masked from 'package:dplyr':
##
##   between, first, last
```

```
long <- melt(setDT(knew.g), id.vars = c("S_ID","StimType"), variable.name = "time")

m2<- lmer(value ~ StimType*time + (1|S_ID), data=long)
Anova(m2)
```

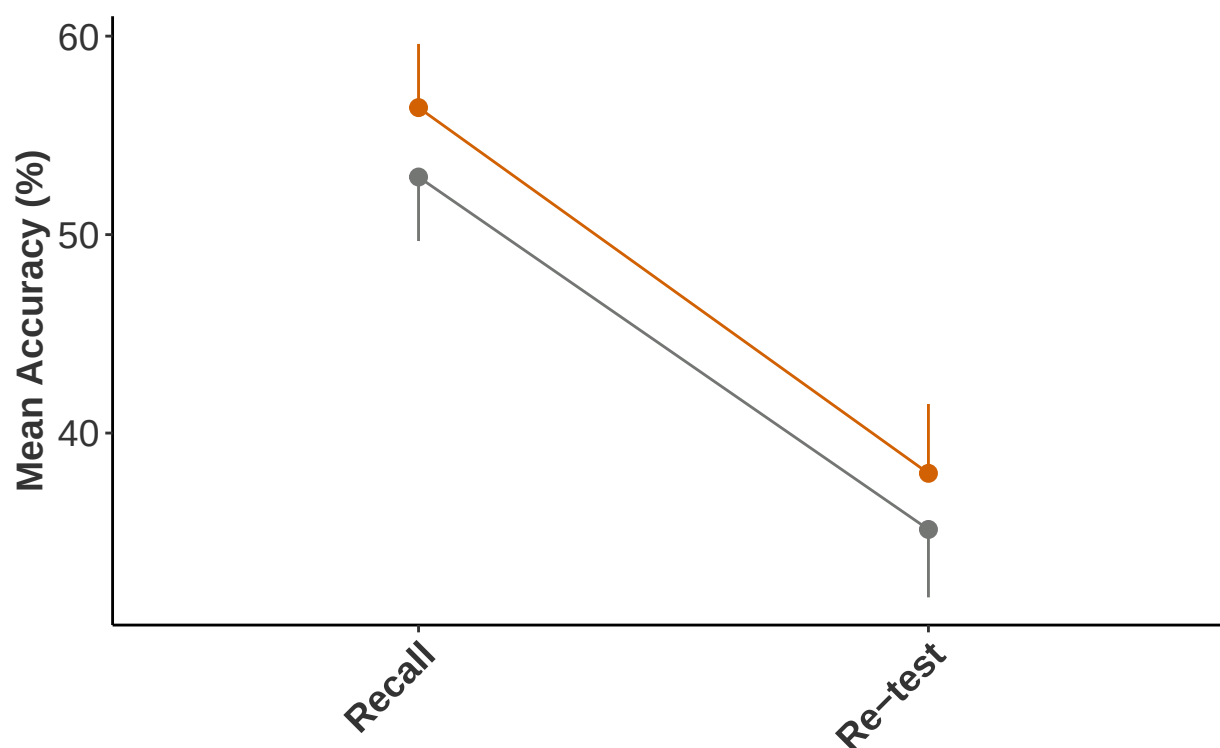
```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: value
##              Chisq Df Pr(>Chisq)
## StimType      6.9045  1  0.008598 **
## time        226.0266  1  < 2.2e-16 ***
## StimType:time  0.0774  1  0.780844
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Plot Recall and Re-test mean accuracy for pairs correctly encoded first time around
df2 <- long %>%
  group_by(StimType, time) %>%
  summarise(acc.mean = (mean(value)/192)*100,
            acc.se = (se(value)/192)*100)
```

```
## 'summarise()' has grouped output by 'StimType'. You can override using the
## '.groups' argument.
```

```
p.Acc.t <- ggplot(subset(df2, StimType=="TI 1:3"), aes(x=time, y=acc.mean, group=StimType, color=StimType)) +
  geom_line(color = "#D16103") +
  geom_pointrange(aes(ymax=acc.mean+acc.se, ymin=acc.mean), color="#D16103") +
  geom_line(data = subset(df2, StimType=="Sham"), color = "#747674") +
  geom_pointrange(data = subset(df2, StimType=="Sham"), aes(ymax=acc.mean, ymin=acc.mean-acc.se), color="#747674") +
  scale_y_continuous(name = "Mean Accuracy (%)") +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1),
        axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
        axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        panel.background = element_blank(),
        legend.title=element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  scale_x_discrete(name = "",
                  labels=c("t1" = "Recall", "t2" = "Re-test")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 20),
        legend.text = element_text(size=14),
        legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("TI 1:3" = "#D16103", "Sham" = "#747674"))

p.Acc.t
```



```
ggsave(p.Acc.t, filename = "Figure_5C.pdf", width = 5, height = 9.2, dpi = 300, units = "cm", device =
```

```
## Fate of all answers from recall to re-test
retest$type_transition <- as.factor(retest$type_transition)
retest$type_transition <- relevel(retest$type_transition, ref = "T2T")
m.acc <- nnet::multinom(type_transition ~ StimType, data = retest)
```

```
## # weights: 15 (8 variable)
## initial value 12162.522304
## iter 10 value 10634.001383
## final value 10623.554569
## converged
```

```
summary(m.acc)
```

```
## Call:
## nnet::multinom(formula = type_transition ~ StimType, data = retest)
##
## Coefficients:
## (Intercept) StimTypeTI 1:3
## CR -1.3267278 -0.14332641
## Never -0.1237968 -0.14512128
```

```
## T2D      -1.5392543      -0.05149545
## T2F      -1.2352561      -0.03230470
##
## Std. Errors:
##      (Intercept) StimTypeTI 1:3
## CR      0.05801164      0.08282168
## Never    0.03878710      0.05488441
## T2D      0.06320719      0.08862090
## T2F      0.05597200      0.07814206
##
## Residual Deviance: 21247.11
## AIC: 21263.11
```

```
exp(coef(m.acc))
```

```
##      (Intercept) StimTypeTI 1:3
## CR      0.2653441      0.8664712
## Never    0.8835593      0.8649174
## T2D      0.2145410      0.9498080
## T2F      0.2907603      0.9682115
```

```
# Check the Z-score for the model (wald Z)
z <- summary(m.acc)$coefficients/summary(m.acc)$standard.errors
z
```

```
##      (Intercept) StimTypeTI 1:3
## CR      -22.870027      -1.7305422
## Never    -3.191701      -2.6441255
## T2D      -24.352520      -0.5810757
## T2F      -22.069180      -0.4134100
```

```
p <- (1 - pnorm(abs(z), 0, 1))*2
p
```

```
##      (Intercept) StimTypeTI 1:3
## CR      0.000000000      0.083533445
## Never    0.001414377      0.008190228
## T2D      0.000000000      0.561189468
## T2F      0.000000000      0.679306277
```

```
Anova(m.acc)
```

```
## # weights: 10 (4 variable)
## initial value 12162.522304
## iter 10 value 10627.728427
## iter 10 value 10627.728417
## final value 10627.728417
## converged

## Analysis of Deviance Table (Type II tests)
##
```

```
## Response: type_transition
##          LR Chisq Df Pr(>Chisq)
## StimType  8.3477  4   0.07964 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m.acc)
```

```
## Call:
## nnet::multinom(formula = type_transition ~ StimType, data = retest)
##
## Coefficients:
##      (Intercept) StimTypeTI 1:3
## CR      -1.3267278      -0.14332641
## Never   -0.1237968      -0.14512128
## T2D     -1.5392543      -0.05149545
## T2F     -1.2352561      -0.03230470
##
## Std. Errors:
##      (Intercept) StimTypeTI 1:3
## CR      0.05801164      0.08282168
## Never   0.03878710      0.05488441
## T2D     0.06320719      0.08862090
## T2F     0.05597200      0.07814206
##
## Residual Deviance: 21247.11
## AIC: 21263.11
```

```
emmeans(m.acc, pairwise ~ StimType | type_transition, adjust="tukey", mode = "prob")
```

```
## $emmeans
## type_transition = T2T:
## StimType  prob      SE df lower.CL upper.CL
## Sham      0.3768 0.00790  8  0.3585  0.3950
## TI 1:3     0.4033 0.00796  8  0.3850  0.4217
##
## type_transition = CR:
## StimType  prob      SE df lower.CL upper.CL
## Sham      0.1000 0.00489  8  0.0887  0.1113
## TI 1:3     0.0927 0.00471  8  0.0819  0.1036
##
## type_transition = Never:
## StimType  prob      SE df lower.CL upper.CL
## Sham      0.3329 0.00768  8  0.3152  0.3506
## TI 1:3     0.3082 0.00749  8  0.2909  0.3255
##
## type_transition = T2D:
## StimType  prob      SE df lower.CL upper.CL
## Sham      0.0808 0.00444  8  0.0706  0.0911
## TI 1:3     0.0822 0.00446  8  0.0719  0.0925
##
## type_transition = T2F:
## StimType  prob      SE df lower.CL upper.CL
```

```
## Sham      0.1095 0.00509 8 0.0978 0.1213
## TI 1:3    0.1135 0.00515 8 0.1017 0.1254
##
## Confidence level used: 0.95
##
## $contrasts
## type_transition = T2T:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:3 -0.02656 0.01122 8 -2.368 0.0454
##
## type_transition = CR:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:3 0.00724 0.00679 8 1.067 0.3172
##
## type_transition = Never:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:3 0.02467 0.01073 8 2.298 0.0506
##
## type_transition = T2D:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:3 -0.00136 0.00629 8 -0.215 0.8349
##
## type_transition = T2F:
## contrast      estimate      SE df t.ratio p.value
## Sham - TI 1:3 -0.00400 0.00724 8 -0.552 0.5963
```

```
g_bdata <- ddply(retest, c("S_ID", "StimType", "type_transition"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/192,
  meanRespTypePerc = meanRespType*100,
  medianRT = median(RT, na.rm=TRUE),
  meanRT = mean(RT, na.rm=TRUE)
)
```

```
level_order <- c('T2T', 'T2F', 'T2D', 'CR', 'Never')
```

```
label_name <- c('Target->Target', 'Target->Foil', 'Target->Distractor', 'Incorrect->Target', 'Incorrect->Foil')
```

```
p.Transi <- ggplot(g_bdata, aes(x = type_transition, y = meanRespTypePerc, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.5) +
  scale_y_continuous(name = "Mean Response (%)",
    breaks = seq(0, 100, 20),
    limits=c(0, 100)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
```

```

scale_x_discrete(limits = level_order,
                 breaks=level_order,
                 labels=label_name) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 20),
        legend.text = element_text(size=14),
        legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("

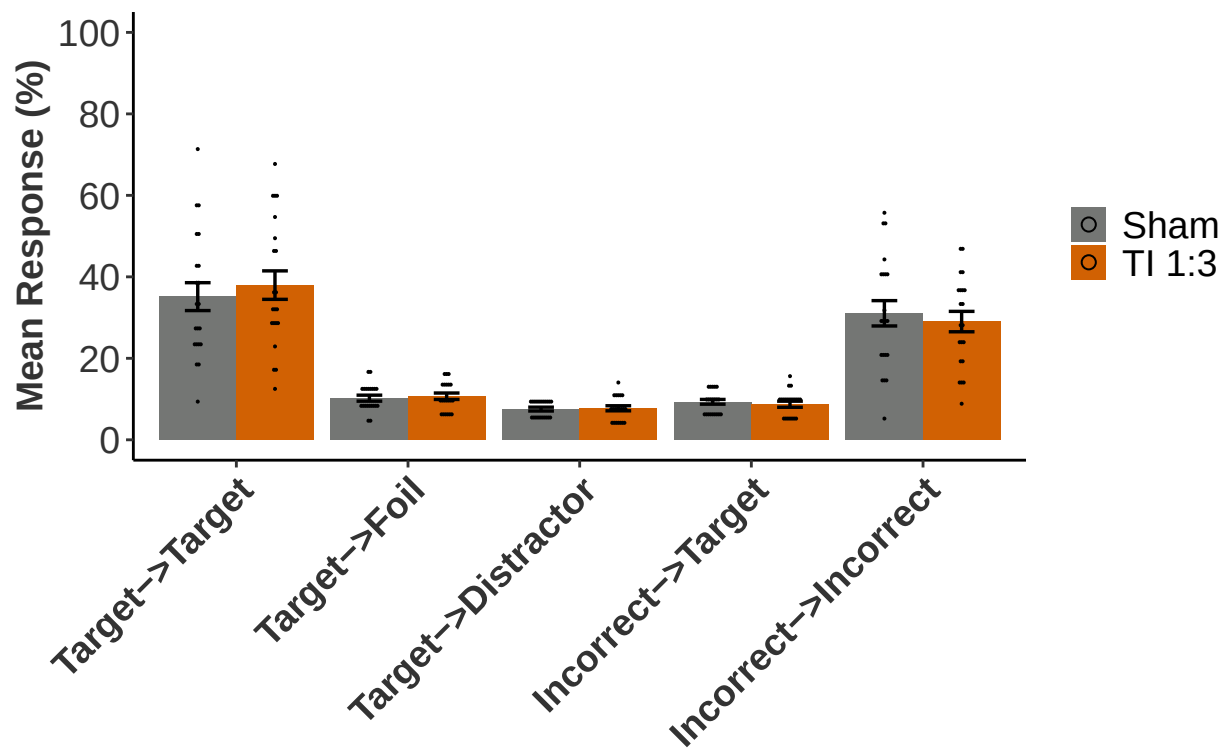
```

p.Transi

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

```



```

ggsave(p.Transi, filename = "Figure_S8C.tiff", width = 11, height = 12, dpi = 300, units = "cm", device

```

```

## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.

```

```

##
## RT for correct trials

```

```
recall_correct <- subset(recall, Accuracy=="1")
df.ir <- recall_correct %>%
  group_by(StimType, S_ID) %>%
  summarise(t1 = median(CorrectRT))
```

Recall - Re-test RT

'summarise()' has grouped output by 'StimType'. You can override using the
'.groups' argument.

```
retest_correct <- subset(retest, Accuracy=="1")
df.lr <- retest_correct %>%
  group_by(StimType, S_ID) %>%
  summarise(t2 = median(CorrectRT))
```

'summarise()' has grouped output by 'StimType'. You can override using the
'.groups' argument.

```
RTall <- merge(df.ir, df.lr, by=c("S_ID", "StimType"))
long <- melt(setDT(RTall), id.vars = c("S_ID", "StimType"), variable.name = "time")

m2 <- lmer(value ~ StimType*time + (1|S_ID), data=long)
Anova(m2)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: value
##           Chisq Df Pr(>Chisq)
## StimType    0.0001  1    0.9905
## time        0.6466  1    0.4213
## StimType:time 0.0065  1    0.9355
```

```
df2 <- long %>%
  group_by(StimType, time) %>%
  summarise(mean = mean(value),
            se = se(value))
```

'summarise()' has grouped output by 'StimType'. You can override using the
'.groups' argument.

```
p.RT.t <- ggplot(subset(df2, StimType=="TI 1:3"), aes(x=time, y=mean, group=StimType, color=StimType, p
  geom_line( color = "#D16103") +
  geom_pointrange(aes(ymax=mean+se, ymin=mean), color="#D16103") +
  geom_line(data = subset(df2, StimType=="Sham"), color = "#747674") +
  geom_pointrange(data = subset(df2, StimType=="Sham"), aes(ymax=mean, ymin=mean-se), color = "#747674"),
  scale_y_continuous(name = "Median Target RT (s)",
    breaks=c(seq(3,4,0.5)), limits = c(3, 4)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1)
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
```

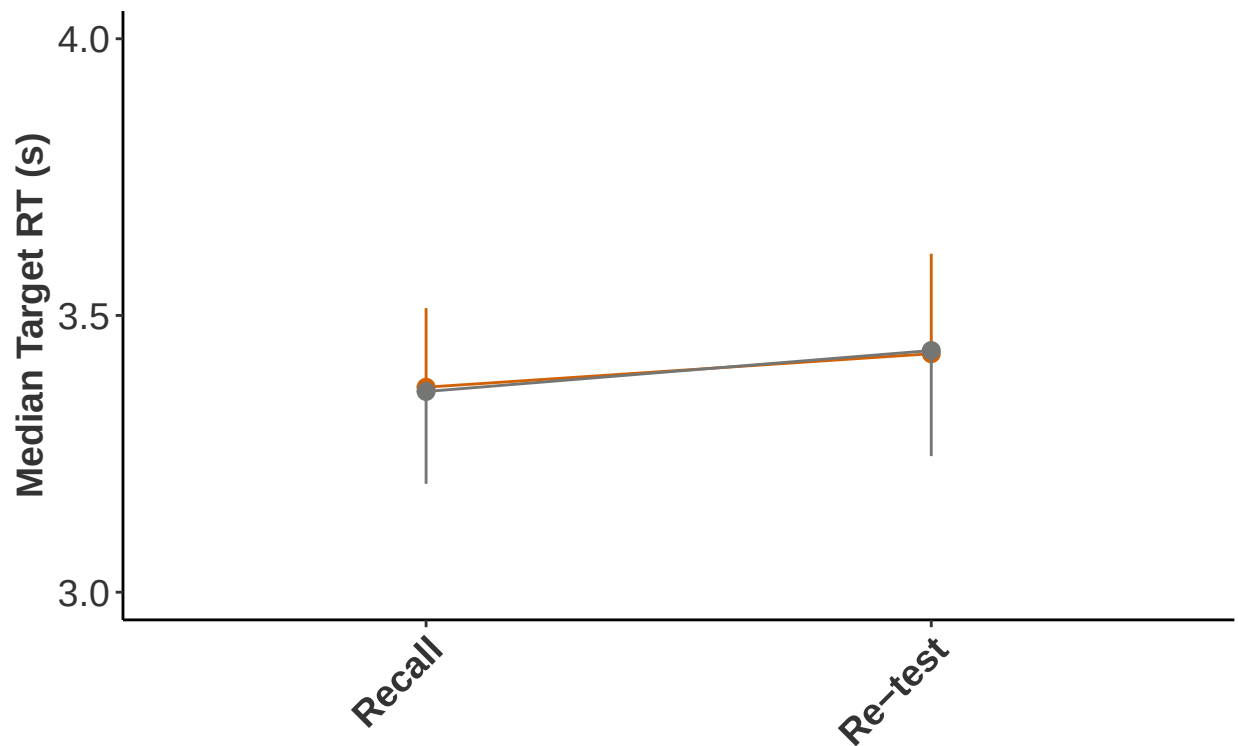


```

axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
  legend.text = element_blank(),
  legend.background = element_blank(),
  legend.key=element_blank(),
  panel.background = element_blank(),
  legend.title=element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
scale_x_discrete(name = "",
  labels=c("t1" = "Recall", "t2" = "Re-test")) +
theme(strip.text.x = element_text(size = 14),
      strip.text.y = element_text(size = 20),
      legend.text = element_text(size=14),
      legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("

```

p.RT.t



```

ggsave(p.RT.t, filename = "Figure_5E.pdf", width = 5, height = 9.2, dpi = 300, units = "cm", device =

```

Stimulation Duration

```

# Import data
data_raw <- read_csv("FNAT_stimduration.csv")

# Assign appropriate Stim factors

```

```

data_raw$StimType <- factor(data_raw$StimType)
levels(data_raw$StimType) <- c("Sham", "TI 1:3")

# Assign factors
data_raw$StimType <- factor(data_raw$StimType)
data_raw$S_ID <- factor(data_raw$S_ID)
data_raw$Run <- factor(data_raw$Run)

## Check how long it took participants to respond for Run 1 and Run 2

data_raw %>% group_by(StimType, Run) %>%
  summarise(mean = mean(stimdur_min),
            sd = sd(stimdur_min), min = min(stimdur_min), max = max(stimdur_min))

```

```

## # A tibble: 4 x 6
## # Groups:   StimType [2]
##   StimType Run    mean    sd   min   max
##   <fct>    <fct> <dbl> <dbl> <dbl> <dbl>
## 1 Sham      1     11.4  1.18  9.11  14.3
## 2 Sham      2      8.23  1.70  5.33  12.3
## 3 TI 1:3    1     11.5  1.05  9.34  14.2
## 4 TI 1:3    2      8.19  1.60  4.89  11.9

```

```

# Select only stimulation and Run 1 -- only run 1 got stimulation
TIIdur <- subset(data_raw, StimType=="TI 1:3" & Run=="1")

# Descriptives for duration - average over subject

g_dur <- ddply(TIIdur, c("S_ID"), summarise,

              stimdur_min = sum(stimdur_min, na.rm=TRUE))

g_dur %>% summarize(mean = mean(stimdur_min),
                  sd = sd(stimdur_min), min = min(stimdur_min), max = max(stimdur_min))

```

```

##      mean      sd      min      max
## 1 34.52659 3.002233 29.37287 40.35236

```

```

# Correlate the duration of stimulation with accuracy in the recall during stimulation

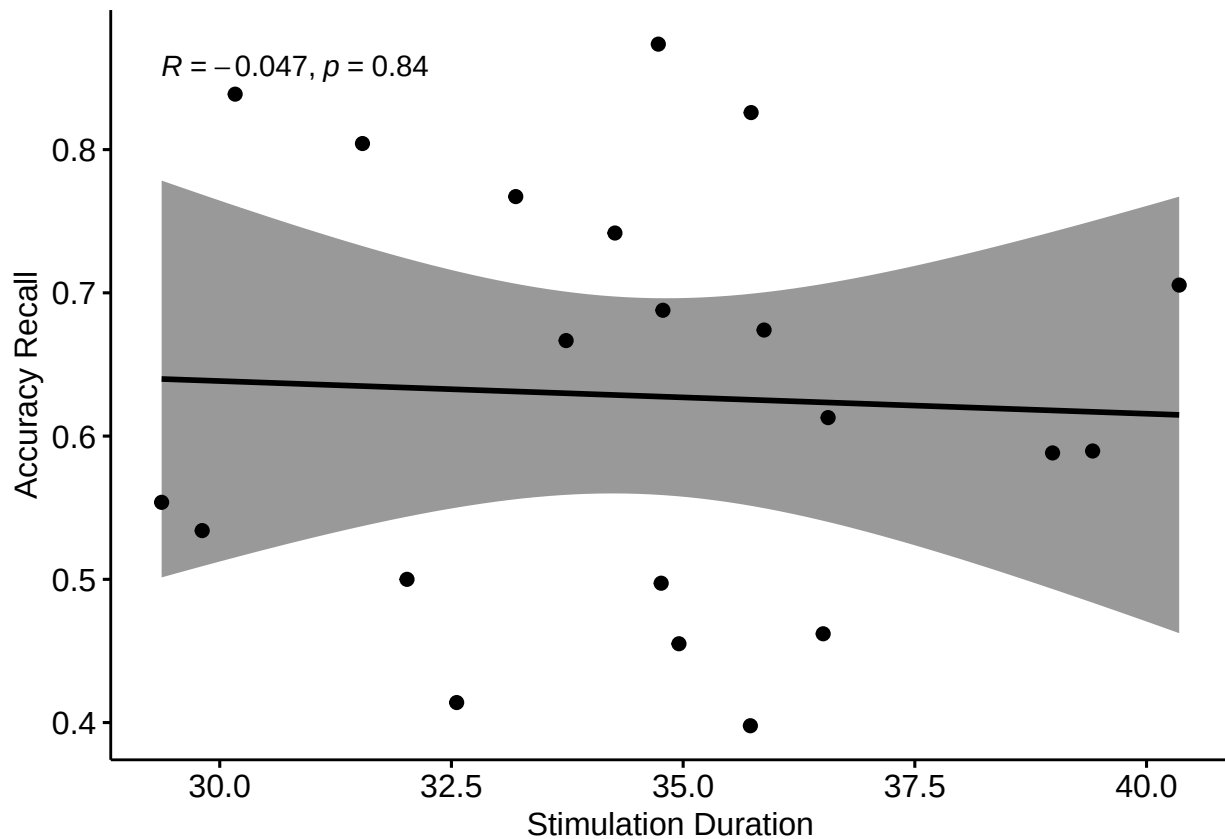
g_recall_TI <- ddply(subset(recall, StimType=="TI 1:3"), c("S_ID"), summarise,
                    mean_acc_recall = mean(Accuracy),
                    mean_acc_perc_recall = mean_acc_recall*100
)

total <- merge(g_recall_TI, g_dur, by=c("S_ID"))

library("ggpubr")
ggscatter(total, x = "stimdur_min", y = "mean_acc_recall",
          add = "reg.line", conf.int = TRUE,

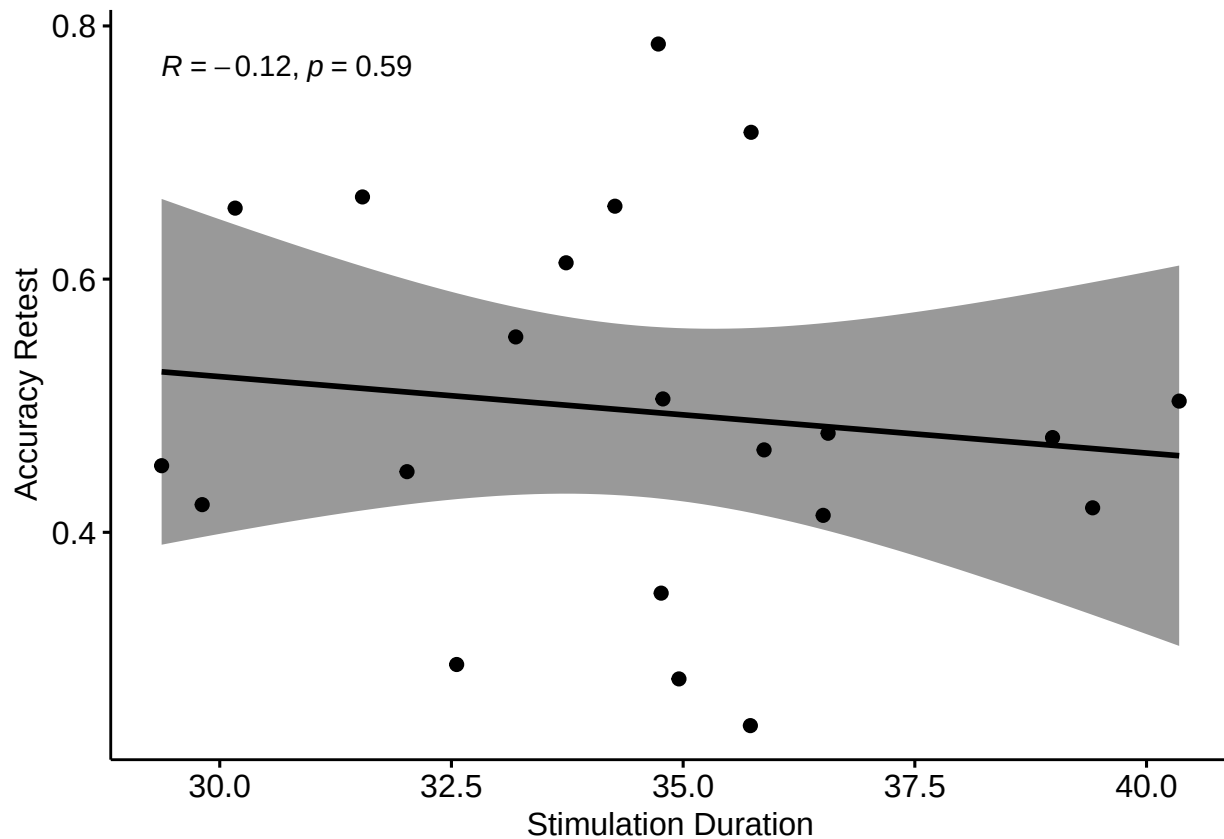
```

```
cor.coef = TRUE, cor.method = "pearson",
xlab = "Stimulation Duration", ylab = "Accuracy Recall")
```



```
# Correlate the duration of stimulation with accuracy in the re-test
g_retest_TI <- ddply(subset(retest, StimType=="TI 1:3"), c("S_ID"), summarise,
  mean_acc_retest = mean(Accuracy),
  mean_acc_perc_retest = mean_acc_retest*100
)
total <- merge(g_retest_TI, g_dur, by=c("S_ID"))

ggscatter(total, x = "stimdur_min", y = "mean_acc_retest",
  add = "reg.line", conf.int = TRUE,
  cor.coef = TRUE, cor.method = "pearson",
  xlab = "Stimulation Duration", ylab = "Accuracy Retest")
```



```
g_dur %>% summarize(mean = mean(stimdur_min+10),
                    sd = sd(stimdur_min+10), min = min(stimdur_min+10), max = max(stimdur_min+10))
```

```
##      mean      sd      min      max
## 1 44.52659 3.002233 39.37287 50.35236
```

```
# Sum task durations
```

```
g.durations <- ddply(data_raw, c("S_ID", "StimType", "Run"), summarise,
                    sum = sum(stimdur_min),
                    std = sd(stimdur_min)
)
```

```
g.durations %>% group_by(Run) %>%
  summarise(mean = mean(sum),
            sd = sd(sum), min = min(sum), max = max(sum))
```

```
## # A tibble: 2 x 5
##   Run  mean    sd   min   max
##   <fct> <dbl> <dbl> <dbl> <dbl>
## 1 1      34.3  3.18  28.5  42.1
## 2 2      16.4  3.23  10.5  24.0
```